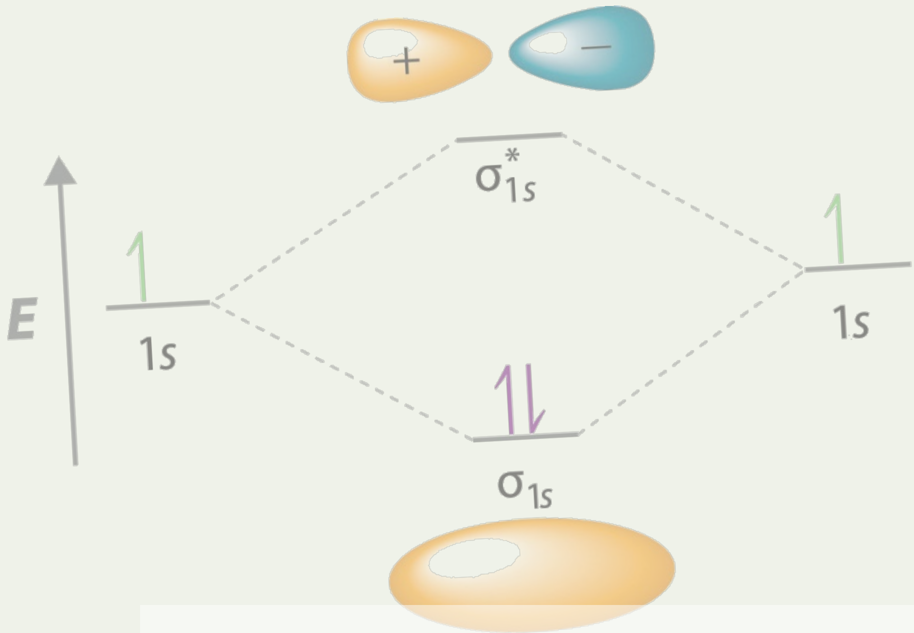

Condensed Matter Theory

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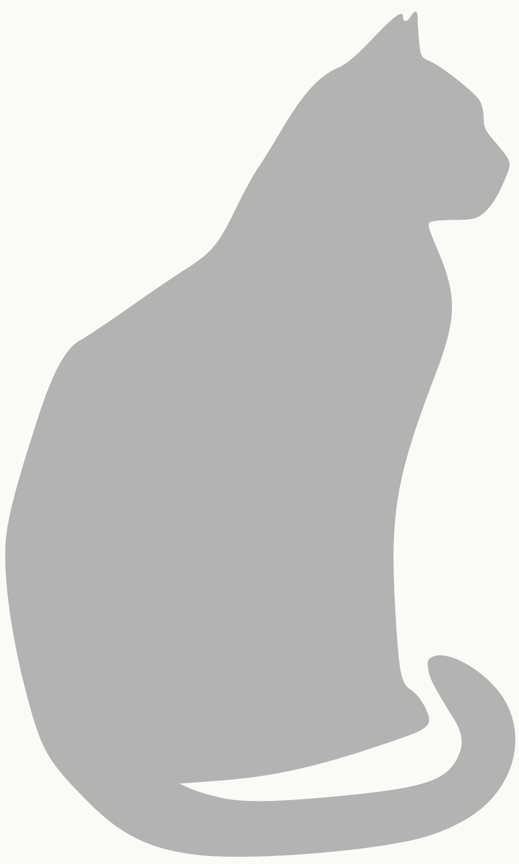

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Chapter 1 Preparation

1.1 Outline

Free electron gas Drude model; Sommerfeld model; magneto-transport

Electrons in a crystal Crystal structure (such as symmetry); Diffraction experiments; Lattice vibrations

Band theory Bloch's theorem; Nearly free electron model (simplified model); Tight binding Model (simplified model); Applications of band theory; Semiconductor devices

Exotic band theory Dirac band; [Topological band; Flat band]

Beyond band theory (Many-body physics) Magnetism; Superconductivity

1.2 Course Overview

Lecture 1 Introduction

Lecture 2 Free electron model of metals

Lecture 3 Lattice vibrations and phonons

Lecture 4 Crystal symmetry and crystal structure

Lecture 5 Scattering experiments (X-ray diffraction and electron diffraction)

Lecture 6 Electron waves in crystals (Bloch's theorem)

The two models help us to Understand why there's a gap in some crystal

Lecture 7 Band theory I: Nearly free electron model

Lecture 8 Band theory II: Tight binding model

Lecture 9 Band theory III: Fermi surface ad effective mass

Lecture 10 Physics of semiconductors

Lecture 11 Electrons in a magnetic field: Quantum Hall effect

Lecture 12 Dirac band theory I: band structure of graphene

Lecture 13 Dirac band theory II: Landau quantization of graphene

Lecture 14 Magnetism

Lecture 15 Superconductivity

Lecture 16 Topological Physics

1.3 References

- Introduction to Solid State Physics, Charless Kittel
- Solid State Physics, Neil Ashcroft and David Mermin
- The Oxford Solid State Basics, Steven H. Simon
- Solid State Physics: Essential Concepts, David W. Snoke

Chapter 2

Chapter 3 Free electron model of metals

3.1 Durde model (1900) for electrons in metals

3.2 Ohm's Law in macroscope

$$V = \int_a^b E\,\mathrm{d}x = Ed,\ V = IR \tag{3.1}$$

Define current density $j = I/A$, therefore

$$Ed = jAR,\ E = \left(\frac{RA}{d}\right)j$$

then, we can define the conductivity $\sigma = 1/\rho$, where $\rho = RA/d$ is the resistivity. So,

$$\boldsymbol{j} = \sigma \boldsymbol{E} \tag{3.2}$$

3.3 Resistivity

1. Metals:
   - Very conductive
   - The resistivity increases with increasing temperature
2. Semiconductors
   - Limited conductance
   - The resistivity decreases with increasing temperature
3. Insulators: Very weak conductance

3.4 Summerfeld model (1920s) for electrons in metals

3.5 Drude model

Without electric field:

- Free electron moving randomly
- $\frac{1}{2}mv_{\mathrm{rms}}^2 = \frac{3}{2}k_BT$, $v_{\mathrm{rms}} = 1.2 \times 10^5$ m/s at room temperature.

With elecric field, the drift velocity $v_{\mathrm{d}} \ll v_{\mathrm{rms}}$, $v_d \sim 10^{-5}\,\mathrm{m/s}$. The definition of the drift velocity is $v_{\mathrm{d}} = l/\tau$.
Let τ be the scattering time and $1/\tau$ be the scattering rate. Consider in a time interval $\mathrm{d}t$, a fraction $\mathrm{d}t/\tau$ of th electrons are scattered, and their average velocity becomes zero.

3.6 Classical Hall effet

3.7 Drude model with a magnetic field

Consider Drude model with applied electric and magnetic fields:

$$m^* \dot{\boldsymbol{v}} = -e(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) - m^* \boldsymbol{v}/\tau$$

In steady state, we have

$$\boldsymbol{E} = -\frac{m^* \boldsymbol{v}}{e\tau} - \boldsymbol{v} \times \boldsymbol{B}$$

and we can rewritten it into 3 components:

$$\begin{aligned} \frac{v_x}{\tau} - \omega_c v_y &= -\frac{e}{m} E_x \\ \frac{v_y}{\tau} + \omega_c v_x &= -\frac{e}{m} E_y \\ \frac{v_z}{\tau} &= -\frac{e}{m} E_z \end{aligned}$$

then we arrive at

$$\begin{aligned} v_x &= -\frac{e\tau}{m(1+\omega_c^2\tau^2)}(E_x + \omega_c \tau E_y) \\ v_y &= -\frac{e\tau}{m(1+\omega_c^2\tau^2)}(E_y - \omega_c \tau E_y) \\ v_z &= -\frac{e\tau}{m} E_z \end{aligned}$$

and we can write it into matrix form

$$\begin{pmatrix} v_x \\ v_y \\ v_z \end{pmatrix} = \hat{\sigma} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix}$$

or $\hat{\rho} = \sigma^{-1}$.

3.8 Summerfeld Model

3.8.1 Electron wave

Time-independent Schrödinger equation

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\boldsymbol{r}) + U\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r}) \tag{3.3}$$

For a free electron: $U = 0$,

$$\nabla^2\psi(\boldsymbol{r}) = -\frac{2mE}{\hbar^2}\psi(\boldsymbol{r})$$

The solution is a plane wave

$$\psi(\boldsymbol{r}) = \frac{1}{\sqrt{V}}\,\mathrm{e}^{\mathrm{i}(k_x x + k_y y + k_z z)} \tag{3.4}$$

Energy

$$E = \frac{\hbar^2 k^2}{2m} = \frac{\hbar^2(k_x^2 + k_y^2 + k_z^2)}{2m} \tag{3.5}$$

apply the Born-von Karman periodic boundary conditions

$$\begin{aligned}
\psi(x + L_x, y, z) &= \psi(x, y, z)\\
\psi(x, y + L_y, z) &= \psi(x, y, z)\\
\psi(x, y, z + L_z) &= \psi(x, y, z)
\end{aligned} \tag{3.6}$$

States in k-space

$$k_x = \frac{2\pi n_1}{L_x},\ k_y = \frac{2\pi n_2}{L_y},\ k_z = \frac{2\pi n_3}{L_z} \tag{3.7}$$

with $n_i \in \mathbb{Z}$. In each unit (3D grid) with size d^3: $\Delta d = \frac{2\pi}{L_x}[(x+1) - (-x)]$. The volume is $\frac{2\pi}{L_x} \cdot \frac{2\pi}{L_y} \cdot \frac{2\pi}{L_z}$, there are $2 \times V/(2\pi)^3$ quantum states per unit volume in k-space, while the density is $\frac{2}{V} = 2 \times \frac{1}{\left(\prod_i \frac{2\pi}{L_i}\right)}$. One can simply think that the allowed states inside a sphere. The volume of the sphere should be

$$V = \frac{4}{3}\pi k_F^3$$

while the density is

$$2 \times \frac{1}{(\frac{2\pi}{L})^3}$$

then, the number of electrons in the sphere should be

$$N = 2 \times \frac{(4\pi/3)k_F^3}{(2\pi/L)^3} = \frac{k_F^3}{3\pi^2}V$$

We define the *Fermi wave vector*: $k_F = (3\pi^2 n)^{1/3}$, which we can compute it from the carrier density $n = N/V$.

3.9 Bosons and Fermions at $T \neq 0$

Bosoons:

$$\bar{n}_i = \frac{g_i}{\exp} \tag{3.8}$$

at $T = 0$,

$$f(E) = \begin{cases} 0, & \text{for } E > E_f \\ 1, & \text{for } E < E_f \end{cases}$$

3.10 Fermi-Dirac Distribution

$$f(E) = \frac{1}{1 + \exp[(E - E_f)/k_B T]} \tag{3.9}$$

3.11 Density of states

$$n(T) = \int f(E) g(E) \,\mathrm{d}E$$

$g(E)$: how many states are allowed at energy E.

$$g(E) = \frac{1}{V} \frac{\mathrm{d}N}{\mathrm{d}E} \tag{3.10}$$

we cancalculate the total number

$$N = \frac{k^3}{3\pi^2} V \tag{3.11}$$

and

$$E = \frac{\pi^2 k^2}{2n} \tag{3.12}$$

Subsititute E and N into $g(E) = \frac{1}{V} \frac{\mathrm{d}N}{\mathrm{d}E}$, we can have

$$g(E) = \frac{mk}{\hbar^2 \pi^2} \tag{3.13}$$

which $k = \frac{\sqrt{2mE}}{\hbar}$.

(a) $g_{3D}(E) = \frac{m^*}{\pi^2 \hbar^3} \sqrt{2m^* E} \propto E^{1/2}$

(b) $g_{2D}(E) = \frac{m^*}{\pi \hbar^2} = \text{Const}$

(c) $g_{1D}(E) = \frac{1}{\pi \hbar} \sqrt{\frac{2m^*}{E}} \propto E^{-1/2}$

(d) $g_{0D}(E) = 2\delta(E)$

The total number of electron at $T > 0$ per volume

$$\frac{N}{V} = \int \underbrace{f(E)}_{probability} \cdot \underbrace{g(E) \,\mathrm{d}E}_{\text{can be occupied}}$$

and the energy for the electrons at the Fermi energy level (internal energy) is

$$\bar{E} = \int E f(E) g(E) \,\mathrm{d}E \tag{3.14}$$

to calculate the velocity

$$v(E) = \int v(E) f(E) g(E) \,\mathrm{d}E, \quad \text{and} \quad v(k) = \int v(k) f(k) g(k) \,\mathrm{d}k$$

remember to add a $\frac{V}{(2\pi)^3}$ factor when calculate in k-space. Then, the current density is

$$\boldsymbol{j} = -ne\boldsymbol{v} = -2e \int \frac{\mathrm{d}^3 \boldsymbol{k}}{(2\pi)^3} f(\boldsymbol{k}) v(\boldsymbol{k}) \tag{3.15}$$

Consider $\boldsymbol{j} = \sigma \boldsymbol{E}$, when $\boldsymbol{E} = \boldsymbol{0}$ then $\boldsymbol{j} = \boldsymbol{0}$. Now, check it. At zero field, for every occupied state $\boldsymbol{k}$, there is a state $-\boldsymbol{k}$ occupied with the same probability and the corresponding velocity is same but negative. For the current density in an applied electric field,

$$\boldsymbol{j} = -nev = -2e \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3} f\left(\boldsymbol{k} + \frac{e\tau}{\hbar}\boldsymbol{E}\right) v(\boldsymbol{k}) = -2e \int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3} f(\boldsymbol{k}) v\left(\boldsymbol{k} - \frac{e\tau}{\hbar}\boldsymbol{E}\right) \tag{3.16}$$

since $v(\boldsymbol{k}) = h\boldsymbol{k}/m$

$$\boldsymbol{j} = \frac{e^2\tau}{m}\left[2\int \frac{\mathrm{d}^3\boldsymbol{k}}{(2\pi)^3} f(\boldsymbol{k})\right]\boldsymbol{E} = \frac{e^2\tau}{m} n\boldsymbol{E} = \sigma\boldsymbol{E}$$

which $\sigma = ne\tau^2/m$ is the same as that in Drude mode.

3.11.1 Heat capacity of an electron gas

The change of internal energy when heated from $0K$ to $T > 0$

$$\Delta U = U(T) - U(0) = \int - \int \tag{3.17}$$

- At Mediate temperature: T^3
- At Low temperature: T
- At High temperature: Const

At low temperature, f choose to be step function, and obviously, $\frac{\partial}{\partial T} f$ will be like a δ-function.

3.11.2 Thermal transport

3.12 Limitation of Free electron gas model

Chapter 4 Lattice Vibrations and Phonons

4.1 Specific head capacity in solids

$$m\frac{\mathrm{d}^2u_n}{\mathrm{d}t^2} = \beta(u_{n+1} + u_{n-1} - 2u_n) \tag{4.1}$$

should have wave-like solutions

$$u_n = A\,\mathrm{e}^{\mathrm{i}(\omega t - nak)}$$

Subsititute it to the ODE, then we have the dispersion relation

$$\omega = 2\sqrt{\frac{\beta}{m}}\left|\sin\left(\frac{ka}{2}\right)\right|$$

- The dispersion relation is perodic in k, *with period* $2\pi/a$
- ...

$k = n\frac{2\pi}{Na}$ For a n-number of particles, we have n phonons, each phonon has a vibration, that is a ω.
For small ka, $\omega \approx a\sqrt{\frac{\beta}{m}}|k|$.
$k \in [-\pi/a, \pi/a]$, $k \to k + 2\pi/a$
$\hbar\frac{\pi}{a}\frac{2}{3} \times 3 \to \frac{2\pi}{a}\hbar$
$-\frac{2}{3}\hbar\frac{\pi}{a} \times 3 - \frac{2\pi}{a}\hbar + \frac{2\pi}{a}\hbar \times 2$

4.1.1 A diatomic chain

We have two kind of atoms, and keep the period. Using two kind of equations

$$m\frac{\mathrm{d}^2u_{2n}}{\mathrm{d}t^2} = \beta(u_{2n+1} + u_{2n-1} - 2u_{2n}) \tag{4.2}$$
$$M\frac{\mathrm{d}^2u_{2n+1}}{\mathrm{d}t^2} = \beta(u_{2n+2} + u_{2n} - 2u_{2n+1}) \tag{4.3}$$

Similarly

$$u_{2n} = A\,\mathrm{e}^{\mathrm{i}(\omega t - 2nak)}$$

4.1.2 Gap

For gap: at $k = \left|\frac{\pi}{2a}\right|$, supposed $M > m$, we have

$$\omega_+^2 = \frac{2\beta}{m} > \omega_-^2 = \frac{2\beta}{M}$$

when $m = M$, the gap equals to zero.

Chapter 5 Crystal Symmetry & Crystal Structure

5.1 Crystal Symmetry

5.1.1 Inversion Symmetry

5.1.2 Rotoinversion Symmetry

5.2 Crystal Structure

$$\text{Crystal lattice} + \text{Basis} = \text{Crystal structure} \tag{5.1}$$

Replace a every site (represent the symmetry of crystals) by basis (represent the compound of crystal: one / a group of atom / a molecule / a set of molecule): nvotif
Fourier transfor:

$$\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{R}} = 1,\ A(k) = A(k)\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} \tag{5.2}$$

then $\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}} = 1$.

$$\boldsymbol{a}_i \cdot \boldsymbol{b}_i = 2\pi\delta_{ij} \tag{5.3}$$

Example 5.2.1 (Rec. lattice of BCC)**.**

5.2.1 Structure factor of a monatomic lattice

In BCC structure, there are two atoms in a unit cell: $(0,0,0)$ and $\frac{a}{2}(1,1,1)$. They have the same atomic form factor f

$$S_G = f\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}_1\cdot\boldsymbol{r}_1} + f\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}_2\cdot\boldsymbol{r}_2}$$

where $\boldsymbol{G}_1 = \frac{2\pi}{a}(hx_0 + kx_0 + lx_0)$ and $\boldsymbol{G}_1 \cdot \boldsymbol{r}_1 = 0$; $\boldsymbol{G}_2 = \frac{2\pi}{a}(1 \times \frac{a}{2}h + 1 \times \frac{a}{2}k + 1 \times \frac{a}{2}l)$.

- $S_G = 0$, when $h + k + l$ is an odd integer, there No diffraction.
- $S_G = 2f$, when $h + k + l$ is an even integer, diffraction peak exists.

X-Ray Diffraction: some peeks in I along θ: stands for (100), (111), (110), but (110) is missing. *Missing orders make it easier to identify a particular crystal structure.*
The wave length of the X-Ray source has a range (λ_1, λ_2), corresponging $k_{\max}$ to $k_{\min}$. For each k, we have a circle 'field'.
For incident beam, we can get a series of ring: the distance between the rings $d = a/\sqrt{h^2 + k^2 + l^2}$ for cubic lattice.

5.2.2 Bloch theorem

Lattice vector

$$\boldsymbol{R} = k_1\boldsymbol{x}_1 + k_2\boldsymbol{x}_2 + k_3\boldsymbol{x}_3$$

where k_i should be integer numbers. Bloch electrons

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + U(\boldsymbol{r})\right]\psi(\boldsymbol{r}) = E\psi(\boldsymbol{r})$$

where $U(\boldsymbol{r}+\boldsymbol{R}) = u(\boldsymbol{r})$. The solution should be the form of

$$\psi_k(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u_k(\boldsymbol{r}), \tag{5.4}$$

where $u_k(\boldsymbol{r}) = u_k(\boldsymbol{r}+\boldsymbol{R})$, multiplication of Plane wave and lattice periodic function.

5.2.3 Nearly free electron model

For free electrons, the periodic function

$$E = \frac{\hbar^2 k^2}{2m} \tag{5.5}$$

then, when we apply the Bloch theorem

$$\psi_R(\boldsymbol{r}) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}}u_k(\boldsymbol{r}), \quad u_k(\boldsymbol{r}) = u_k(\boldsymbol{r}+\boldsymbol{R})$$

Example 5.2.2 (Simple cube)**.**

$$E(\boldsymbol{k}) = \epsilon_s - J_0 - 2J_1(\cos(k_x a) + \cos(k_y a) + \cos(k_z a))$$

where

$$-\boldsymbol{k}\cdot\boldsymbol{R} = -k_x\cdot a + ky\cdot 0 + k_z\cdot 0 = -k_x\cdot a$$

In the first Brillouin zone of simple cube, there are several high symmetric points

(a) Γ points: $\boldsymbol{k} = (0,0,0)$

(b) X points: $\boldsymbol{k} = (\pi/a, 0, 0)$

(c) R points: $\boldsymbol{k} = (\pi/a, \pi/a, \pi/a)$

(d) M points: $\boldsymbol{k} = (\pi/a, \pi/a, 0)$

Along different directions, the band structure is different.

$$E(\Gamma) = \epsilon_0 - J_0 - 6J_1, \tag{5.6}$$
$$E(X) = \epsilon_0 - J_0 - 2J_1, \tag{5.7}$$
$$E(R) = \epsilon_0 - J_0 + 6J_1, \tag{5.8}$$

The effective mass. Around the "minimum" point, the electrons behave as the free electrons, i.e.,

$$E = \frac{\hbar^2 k^2}{2m}$$

Since we have (for cubic)

$$E(k) = \epsilon_0 - J_0 - 6J_1 + J_1|\boldsymbol{k}|^2a^2$$

and we can expnad it into Taylor series

$$E = E_0 + a_1\frac{\partial E}{\partial k} + a_2\frac{\partial^2 E}{\partial k^2} + \cdots$$

where $\frac{\partial E}{\partial k}$ around the minimum point. Compare the expressions, we have the effective mass

$$m^* = \frac{\hbar^2}{\partial^2 E/\partial k^2}$$

The bands have antisymmetric properties, i.e.,

$$\frac{\partial E}{\partial k_x} \neq \frac{\partial E}{\partial k_y} \neq \frac{\partial E}{\partial k_z}, \quad \text{and} \quad \frac{\partial^2 E}{\partial k_x^2} \neq \frac{\partial^2 E}{\partial k_y^2} \neq \frac{\partial^2 E}{\partial k_z^2},$$

5.3 Bloch Oscillations

For three dimensional

$$E = \epsilon_0 - J_0 - 2J_1(\cos(k_xa) + \cos(k_ya) + \cos(k_za))$$

For one dimensional

$$E = \epsilon_0 - J_0 - 2J_1\cos(k_xa)$$

5.3.1 Requirements for observation

$$w = \frac{eEa}{\hbar}, \quad T = \frac{2\pi}{eEa} = \frac{h}{eEa}$$

Chapter 6

Chapter 7

Chapter 8 Band theory II: Tight binding model

8.1 Metal, insulator

8.2 Excition

Starding from the Schrödinger equation including column interactions

$$\left[-\frac{\hbar^2}{2m_e^*}\nabla_e^2 - \frac{\hbar^2}{2m_h^*}\nabla_h^2 - \frac{e^2}{4\pi\epsilon\epsilon_0|r_e - r_h|}\right]\psi_{ex} = E\psi_{ex} \tag{8.1}$$

To solve it, extract the eigenvalue E. The Hamiltonian can be separated into two parts: the free electron part and the Hydrogen part

$$E_n^{\text{ex}} = E_g - \frac{m_0^* e^4}{2(4\pi\epsilon\epsilon_0)^2\hbar^2 n^2} = E_g - \frac{R_{\text{ex}}}{n^2} \tag{8.2}$$

where R_{ex} denotes the excition Rydberg.

Chapter 9 Physics of semiconductors

9.1 Intrinsic semiconductors

The number of particles

$$n = \int_{E_c}^{\infty} \mathrm{d}E g(E) f(E) \tag{9.1}$$

where

$$g(E) = \frac{m_n^*}{\pi^2 \hbar^3} \sqrt{2 m_n^* (E - E_c)}$$

Typically, E_f is inside the band gap. In most cases, the energy $E \gg E_f$. So, the distribution function is

$$f(E) = \frac{1}{\mathrm{e}^{(E-E_f)/kT} + 1} \approx \mathrm{e}^{-(E-E_f)/kT}$$

Then, the integral

$$n = \int_{E_c}^{\infty} \frac{m^*}{\pi^2 \hbar^3} \sqrt{2 m^* (E - E_c)}\, \mathrm{e}^{-(E-E_f)/kT}\, \mathrm{d}E = N_c\, \mathrm{e}^{-(E_c - E_f)/kT} \tag{9.2}$$

where $N_c = \frac{2(2\pi m_e^* kT)^{3/2}}{h^3}$.
One can simplify understanding the density of carrier is proportional to

$$n_i^2 = n \cdot p = N_c \cdot N_v\, \mathrm{e}^{-E_k/kT} \tag{9.3}$$

n and p satisfies

$$\begin{cases} n + N_a & = p + N_d, \\ np & = n_i^2 \end{cases}$$

then, we can solve

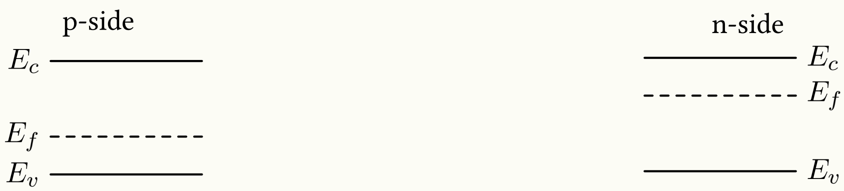
$$n = \frac{N_d - N_a}{2} + \left[\left(\frac{N_d - N_a}{2} \right)^2 + n_i^2 \right]^2, \tag{9.4}$$

$$p = \frac{N_a - N_d}{2} + \left[\left(\frac{N_a - N_d}{2} \right)^2 + n_i^2 \right]^2, \tag{9.5}$$

Case 1 n-type: $N_d - N_a \gg n_i$

Case 2 p-type: $N_d - N_a \ll n_i$

This p-n junction is consist with p-dropped and n-dropped silicon

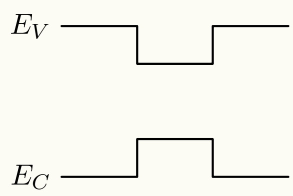


The homojunction: have the same band gap and abnd element.

Behavior of V If apply a positive voltage V, then, build-in potential will reduce; If apply a negative voltage V, then, build-in potential will increase.

Semiconductor heterstructure Connect two different semiconductors together, and the bandgap can be different, also, the band alignment can be different.

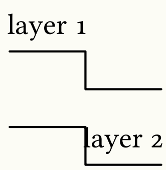
(a) Type I: straddling gap. Application: 2DEG



The generation is AlGaAs, which provide the carier n, and the conducting channel is GaAs, which untdout dopes.

(b) Type II: staggered gap.

For electrons, the electrons locate in layer 2, but for holes, it locate in layer 1, they are separated. They can interact with each other by Coulomb interaction.



layer 1: -
layer 2: +

Without combinations, they can have a longer lifetime.

(c) Type III: broken gap.

Chapter 10 Electrons in a Magnetic Field

Free electrons in a magnetic field

$\longrightarrow$ Bloch electrons (electrons in a periodic potential field) in a magnetic field

For metals, the electrons near the Fermi surface can be considered as the quasifree electrons, where $m^* \to m$. So, we start from the free electrons first.

10.1 Bohr-Sommerfeld Quantization

10.1.1 Classical Hall effect

In the polar coordinate

$$\begin{aligned}
\boldsymbol{r} &= r\cos(\omega t)\boldsymbol{i} + r\sin(\omega t)\boldsymbol{j} \\
\dot{\boldsymbol{r}} &= -r\omega\sin(\omega t)\boldsymbol{i} + r\omega\cos(\omega t)\boldsymbol{j} \\
\ddot{\boldsymbol{r}} &= -r\omega^2\cos(\omega t)\boldsymbol{i} - r\omega^2\sin(\omega t)\boldsymbol{j}
\end{aligned}$$

where the equation of motion

$$m^*\ddot{\boldsymbol{r}} = e\dot{\boldsymbol{r}} \times \boldsymbol{B}$$

If the magnetic field $B = (0, 0, B)$, then

$$m^*(-r\omega^2\cos(\omega t)) = -e \cdot r\omega\cos(\omega t)B$$

where $r = v/\omega \propto \frac{1}{B}$, and the orbit is quantized.
The current density

$$\boldsymbol{j} = \frac{ne^2\tau}{m^*}\boldsymbol{E}$$

while in Drude model, we have

$$\boldsymbol{j} = \hat{\sigma}\boldsymbol{E}$$

where σ is a tensor mentioned in Chapter 1. If we apply the magnetic field, we use

$$\boldsymbol{E} = \frac{m^*}{ne^2\tau}\boldsymbol{j} + \frac{1}{ne}\boldsymbol{j} \times \boldsymbol{B}$$

In the Drude model, the diagonal elements are independent from the magnetic field, while the xy term is linear to the magnetic field.

10.1.2 Bohr-Sommerfeld Quantization rule

The number of wavelength along the trajectory must be integer

$$\frac{1}{\hbar}\oint \boldsymbol{p}\cdot \mathrm{d}\boldsymbol{r} = 2\pi N \tag{10.1}$$

Apply the magnetic field, we need do the replication $\boldsymbol{p}\to\boldsymbol{p}-e\boldsymbol{A}$, hence

$$\frac{1}{\hbar}\oint \boldsymbol{p}\cdot \mathrm{d}\boldsymbol{r} - \frac{1}{\hbar}\oint e\boldsymbol{A}\cdot \mathrm{d}\boldsymbol{r} = 2\pi N \tag{10.2}$$

where $\boldsymbol{p}=\hbar\boldsymbol{k}$. Using the motion equation becomes

$$\hbar\boldsymbol{k} = -e\boldsymbol{r}\times\boldsymbol{B} \tag{10.3}$$

The first term of the integral now becomes

$$\frac{1}{\hbar}\oint \hbar\boldsymbol{k}\cdot \mathrm{d}\boldsymbol{r} = -\frac{e}{\hbar}\oint(\boldsymbol{r}\times\boldsymbol{B})\cdot \mathrm{d}r = -\frac{e}{\hbar}\oint \boldsymbol{B}\cdot(\mathrm{d}\boldsymbol{r}\times\boldsymbol{r}) = -\frac{2e}{\hbar}\oint \boldsymbol{B}\cdot\mathrm{d}\boldsymbol{S} = \frac{2e}{\Phi}$$

and the seconde term

$$\frac{e}{\hbar}\oint \boldsymbol{A}\cdot \mathrm{d}\boldsymbol{r} \xlongequal{\text{Stock' Equation}} \frac{e}{\hbar}\oint(\nabla\times\boldsymbol{B})\cdot \mathrm{d}\boldsymbol{r} = \frac{e}{\hbar}\iint \boldsymbol{B}\cdot\mathrm{d}S = \frac{e}{\hbar}\Phi$$

Hence, the quantization condition becomes

$$\frac{e}{\hbar}\Phi = 2\pi N,\quad \Phi = \frac{2\pi\hbar}{e}N = \frac{h}{e}N = \phi_0 N \tag{10.4}$$

Since $\phi_0 = BS$, so, the area S should be quantized. *If the magnetic field is changed, the radius of motion will change discretely.*

10.1.3 Onsager quantization condition

Going to k-space from r-space for $\boldsymbol{B}=(0,0,B_z)$

$$\hbar|\boldsymbol{k}| = e|\boldsymbol{r}|B,\quad |\boldsymbol{k}| = \frac{|\boldsymbol{r}|}{l_B^2},$$

where $l_B=\sqrt{\frac{\hbar}{eB}}$ denotes the magnetic length. Then, the area

$$S_r = \pi r^2,\quad S_k = \pi k^2, \to S_r = l_B^4 S_k$$

where S_k denotes the area of *Fermi Surface*. Then, the flux

$$\phi = N\phi_0,\quad Bl_B^4 S_k = N\frac{h}{e}$$

Then, we obtain the the quantization condition in k-space

$$l_B^2 S_k = 2\pi N \tag{10.5}$$

To generalized onsager relation

$$l_B^2 S_k = 2\pi\left(N + \frac{1}{2} - \frac{\gamma}{2\pi}\right)$$

Consider a change in B

$$\frac{\hbar}{eB_N} \cdot S_k = 2\pi N, \quad \frac{\hbar}{eB_{N+1}} \cdot S_k = 2\pi(N+1)$$

Hence, the flux area change in k-space

$$S_k \cdot \Delta\left(\frac{1}{B}\right) = \frac{2\pi}{\hbar}e \tag{10.6}$$

10.1.4 Quantization of energy

Starting from

$$l_B^2 S_k = 2\pi\left(N + \frac{1}{2} - \frac{\gamma}{2\pi}\right) \tag{10.7}$$

$$S_k = \pi k^2 \tag{10.8}$$

The energy $E = \frac{\hbar^2 k^2}{2m}$

$$\frac{\hbar}{eB} \cdot \pi \cdot \frac{2mE}{\hbar^2} = 2\pi(N + \frac{1}{2}), \quad E = \hbar\omega\left(N + \frac{1}{2}\right)$$

10.1.5 Full quantum mechanical solution

The Hamiltonian for a free electron

$$\hat{H} = \frac{\hat{p}^2}{2m} \xrightarrow{\boldsymbol{p} \to \boldsymbol{p} - e\boldsymbol{A}} \frac{(\boldsymbol{p} - e\boldsymbol{A})^2}{2m}$$

where $\boldsymbol{B} = \nabla \times \boldsymbol{A}$ and use Landau gauge $\boldsymbol{A} = (0, Bx, 0)$. Since p_y, p_z commutes with the Hamiltonian, it can be replaced with its eigenvalue $\hbar k_y$, $\hbar k_z$. The Schrödinger equation becomes

$$\left[\frac{p_x^2}{2m^*} + \frac{1}{2m^*}(\hbar k_y - eB_x)^2 + \frac{\hbar^2 k^2}{2m^*}\right]\psi = E\psi$$

Rearrange it

$$\left[\frac{p_x^2}{2m^*} + \frac{1}{2m^*}(\hbar k_y - eB_x)^2\right]\psi = \left(E - \frac{\hbar^2 k^2}{2m^*}\right)\psi = E'\psi$$

Denote $x_0 = \frac{\hbar k_y}{eB}$, the Schrödinger equation becomes

$$\left[\frac{p_x^2}{2m^*} + \frac{1}{2}m^*\omega_c^2(x - x_0)^2\right]\psi = E'\psi \tag{10.9}$$

The quantized energy

$$E = \frac{\hbar^2 k_z^2}{2m^*} + \left(N + \frac{1}{2}\right)\hbar\omega_c \tag{10.10}$$

Due to the limitation of boundary,

$$0 < x_0 = \frac{\hbar k_y}{eB} < L_x$$

where $k_y = \frac{2\pi}{L_y} n_y$, $\Delta k_y = \frac{2\pi}{L_y}$, it will be limited within

$$0 < k_y < \frac{eBL_x}{\hbar''}$$

So, the total number of state

$$n = \frac{eBL_x/\hbar}{2\pi/L_y} = \frac{eB(L_x L_y)}{h}$$

and the density of states

$$\text{DOS} = \frac{eB(L_x L_y)/\hbar}{L_x L_y} g_s = \frac{eB}{h} g_s = \phi_0 B g_s \propto B$$

Usually, $g_s = 2$, but for like graphene, $g_s > 2$. By increasing the magnetic field, one can confine all of the electrons within the same energy (such as the ground) state.

10.2 Landau Quantization

10.2.1 Filling factor

With a given E_f, the number of Landau levels filled with states ν.

$$\nu = \frac{n}{eB/h} = \frac{hn_B}{eB} \tag{10.11}$$

The Landau level will broaden due to scattering. For a given scattering time τ_q, the broaden energy

$$\Delta E \sim \frac{\hbar}{\tau_q}$$

due to Heisenberg's uncertainty principle. When $\Delta E > \hbar\omega_c$, or $\frac{\hbar}{\tau_c} > \hbar\omega_c$, $\omega_c\tau_c < 1$. We denotte $\mu = e\tau/m$, then

$$\omega_c \cdot \frac{\mu m^*}{e} = \frac{eB}{m^*}\frac{\mu m^*}{e} < 1, \quad B\mu_q < 1$$

So, when $B\mu_q \geq 1$, we could not distinguish the levels. where μ_q denotes the quantum mobility. E.g, for $B = 1\,\mathrm{T}$, $\mu_q \to 10^3\,\mathrm{cm/N \cdot s}$.

10.3 Shubnikov-de Hass Oscillators

10.3.1 Zeeman splitting

Each Landau level will separate into two levels.

$$E_c = \hbar\frac{eB}{m^*}, \quad E_z = g\mu_B B$$

In case that $E_c < E_z$, then $\hbar eB/m^* < g\mu_B B$, $\hbar eB < gm^*\mu_B B$.
If tune B to B_z and $B_{||}$, then

$$\frac{E_c}{E_B} = \frac{\frac{\hbar eB}{m^*}\cos\theta}{g\mu_B B} = \frac{\hbar e\cos\theta}{gm^*\mu_B}$$

10.4 Integer Quantum Hall Effect

10.4.1 Edge modes

The Hamiltonian with the potention $\phi(x)$

$$\hat{H} = \frac{p_x^2}{2m^*} + \frac{1}{2m^*}(\hbar k_y - eBx)^2 + \phi(x) \tag{10.12}$$

The eigenenergy will be

$$E(\hbar k) = \left(N + \frac{1}{2}\right)\hbar\omega_c + \phi\left(\frac{\hbar k_y}{eB}\right)$$

The current

$$I = -e\int_{-\infty}^{\infty} \frac{\mathrm{d}k}{2\pi} v_y(k) = \frac{e}{2\pi l_B^2}\int_0^x \mathrm{d}x \frac{1}{eB}\frac{\delta\phi}{\delta x} = \frac{e}{2\pi\hbar}\Delta\mu$$

where $\frac{\partial E}{\partial k_y} = \frac{\partial \phi}{\partial x} \cdot \left(\frac{\hbar}{eB}\right)$. In the bulk, the electrons motion counterclockwise; But on the both edges of top and top, the electrons will be "perfectly scattered" (reflected) by the edges and do hafl-circle counterclockwise motion. Obviously, the direction of the currents is different.

10.5 Fractional Quantum Hall Effect

10.5.1 Composite fermions

$$B = n_e\phi_e(2j + 1/\nu^*)$$

Then, $\nu = \frac{\nu^*}{2j\nu^*+1}$.
If consider $j = 1$, which means that each electrons attached to flux, then

$$\nu = \frac{1}{3}, \to \nu^* = 1.$$

10.5.2 Wigner crystal

The electron-electron interaction is so large that forms the lattice. The bonds are Columb interactions. When a strong filed is applied on a level, the electrons will be fixed and form the lattice.

Chapter 11 Dirac Band Theory

11.1 Introduction to Graphene

Why μ in graphene is very high

(a) Dirac band: $m^* = 0$.

(b) Electron-phonon scattering is weak.

(c) τ_i increase: μ_i increase.

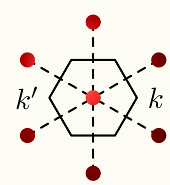
11.1.1 Honeycomb lattice and its Brillouin zone

Two vectors

$$a_1 = \frac{a}{2}(1, \sqrt{3}), \quad a_2 = \frac{a}{2}(1, -\sqrt{3})$$

We want to get the $E(k)$ relation, so we obtain the reciprocal lattice first

$$b_1 = \frac{2\pi}{a}\left(1, \frac{\sqrt{3}}{3}\right), \quad b_2 = \frac{2\pi}{a}\left(-1, \frac{\sqrt{3}}{3}\right).$$



with the first Brillouin zone

$$k = \frac{4\pi}{3a}(1, 0), \quad k' = \frac{4\pi}{3a}(-1, 0).$$

11.1.2 Tight-binding method to calculate band structure of graphene

The Boloch wave function

$$\psi_A = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}_A}^{N} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_A} \varphi_A(\boldsymbol{r} - \boldsymbol{R}_A), \tag{11.1}$$

$$\psi_B = \frac{1}{\sqrt{N}} \sum_{\boldsymbol{R}_B}^{N} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_A} \varphi_B(\boldsymbol{r} - \boldsymbol{R}_B) \tag{11.2}$$

where ψ_A denotes the atomic orbital of A-atom, also for ψ_B. The total wavefunction

$$\psi = \sum_n b_n \psi_n = b_A \psi_A + b_B \psi_b = b_A \frac{1}{\sqrt{N}} \sum_{R_A} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_A} \varphi_A(\boldsymbol{r} - \boldsymbol{R}_A) + b_B \frac{1}{\sqrt{N}} \sum_{R_B} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_B} \varphi_B(\boldsymbol{r} - \boldsymbol{R}_B) \tag{11.3}$$

The eigen function

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

Project all the wavefunction of A-atom

$$\langle\psi_A|\hat{H}|\psi\rangle = E\langle\psi_A|\psi\rangle$$

Then, we have

$$b_A\frac{1}{\sqrt{N}}\sum_{R_A} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_A}\langle\varphi_A|\hat{H}|\varphi_{A'}(\boldsymbol{r}-\boldsymbol{R}_A)\rangle + b_B\frac{1}{\sqrt{N}}\sum_{R_B} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_B}\langle\varphi_A|\hat{H}|\varphi_B(\boldsymbol{r}-\boldsymbol{R}_B)\rangle$$
$$= E\left[b_A\frac{1}{\sqrt{N}}\sum_{R_A} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_A}\langle\varphi_A|\varphi_A(\boldsymbol{r}-\boldsymbol{R}_A)\rangle + b_B\frac{1}{\sqrt{N}}\sum_{R_A} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_B}\langle\varphi_A|\varphi_B(\boldsymbol{r}-\boldsymbol{R}_B)\rangle\right] \quad (11.4)$$

Rearrange it

$$b_A\langle\varphi_A|\hat{H}|\varphi_B\rangle\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_A} + b_B\langle\varphi_A|\hat{H}||\varphi_B\rangle\sum_{R_B}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_B} = Eb_A\,\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_A}$$

Substitute the orthonormality $\langle\varphi_A|\varphi_B\rangle = \delta_{A,B}\epsilon_{A,B}$

$$b_A\epsilon_0 + b_B t\sum_{i=1,2,3}\mathrm{e}^{\mathrm{i}k\cdot k_i} = Eb_A$$

Finally, we arrive at

$$b_A\epsilon_0 + b_B t\sum_i \mathrm{e}^{\mathrm{i}k\cdot\delta_i} = Eb_a, \quad (11.5)$$

Similarly, we can project by $\langle\varphi_B|$

$$b_A t\sum_i \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}_i} + b_B\epsilon = Eb_B \quad (11.6)$$

we define

$$f(k) = \sum_i \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}_i}$$

Then, we can write them into matrix,

$$\begin{pmatrix}\epsilon_0 & tf(k)\\ tf^*(k) & \epsilon\end{pmatrix}\begin{pmatrix}b_A\\ b_B\end{pmatrix} = E\begin{pmatrix}b_A\\ b_B\end{pmatrix}, \mathbf{A}\boldsymbol{b} = E\boldsymbol{b} \quad (11.7)$$

The determine $\det(\mathbf{A}-\lambda\mathbb{1})$ should be zero

$$\begin{vmatrix}\epsilon_0 - E & tf(k)\\ tf^*(k) & \epsilon_0 - E\end{vmatrix} = 0$$

We can obtain the $E(k)$ relation

$$E = \epsilon_0 \pm t|f(k)|. \quad (11.8)$$

Make the replacement

$$\boldsymbol{K} = \frac{4\pi}{3a}(1,0)\to(0,0), \quad \boldsymbol{q} = \boldsymbol{k}-\boldsymbol{K}, \quad \boldsymbol{p} = \hbar\boldsymbol{q}$$

Then

$$f(k) = \mathrm{e}^{\mathrm{i}\boldsymbol{k}_x a/\sqrt{3}} + 2\,\mathrm{e}^{\mathrm{i}-\mathrm{i}k_x a/2\sqrt{3}}\cos(k_x a/2) = \mathrm{e}^{\mathrm{i}q_y a/\sqrt{3}} + 2\,\mathrm{e}^{\mathrm{i}q_y a/2\sqrt{3}}, \quad (11.9)$$

$$f(q) = \cos\left[\frac{(q_x+\frac{4\pi}{3a})a}{2}\right] = \mathrm{e}^{\mathrm{i}q_y a/\sqrt{3}} + 2\,\mathrm{e}^{-\mathrm{i}q_y a/2\sqrt{3}}\left(\cos\frac{q_x a}{2}\cos\frac{4\pi}{3} - \sin\frac{q_x a}{2}\sin\frac{4\pi}{3}\right) \quad (11.10)$$

Expand $f(q)$ to the first order

$$f(q)=\frac{\sqrt{3}a}{2}(\mathrm{i}q_y-q_x)+\mathrm{i}\frac{a^x}{4}q_x\cdot q_y=\frac{\sqrt{3}a}{2}(\mathrm{i}q_y-q_x) \tag{11.11}$$

We can rewrite

$$E(k)=\pm t|f(k)|=\pm t\frac{\sqrt{3}a}{2\hbar}|q|\hbar=v_F|\boldsymbol{p}|$$

One can rewrite the Hamiltonian interms of Pauli matrices

$$H_k(p)=v_F(\sigma_x p_x+\sigma_y p_y)=v_F\boldsymbol{\sigma}\cdot\boldsymbol{p} \tag{11.12}$$

11.2 Calculation about TBC

Using the approximation

(a) Only consider the nearest neighbor hopping

$$\langle\phi_A|H|\phi_B\rangle\neq 0$$

$$\epsilon_0=0,\quad E_\pm=\pm V_0|f(\boldsymbol{k})|$$

(b) $f(\boldsymbol{k})\approx\frac{\sqrt{3}}{2}a(\mathrm{i}q_y-\xi q_x)$.

11.3 Chirality

(a) For conduction band electron: $\boldsymbol{\sigma}\cdot\boldsymbol{n}=1$.

(b) For valence band electron: $\boldsymbol{\sigma}\cdot\boldsymbol{n}=-1$.

11.4 Dirac equation with a mass

The Hamiltonian

$$H_k=\begin{pmatrix}\Delta & -\gamma_0 f(\boldsymbol{k})\\ -\gamma_0 f^*(\boldsymbol{k}) & -\Delta\end{pmatrix}$$

The energy

$$E_\pm=\pm\sqrt{\Delta^2+(\gamma_0|f(\boldsymbol{k})|)^2}=\pm\sqrt{(m_D v_F^2)^2+(\hbar v_F k)^2}$$

where $m_D=\Delta/v_F^2$. The gap is

$$\Delta E=E_+-E_-=2\sqrt{\Delta^2+(\gamma_0|f(\boldsymbol{k})|)^2},$$

and we can open a gap in the band.

11.5 Density of states in Graphene

$$g(E) = g_s g_v vv \frac{|E|}{2\pi\hbar^2 v_F^2}$$

The total carrier density

$$n = \int_0^\infty g(E)f(E)\,\mathrm{d}E = \int_0^{E_f} \frac{g_s g_N E}{2\pi\hbar^2 v_F^2}\,\mathrm{d}E = \frac{g_s g_V E}{2\pi\hbar^2 v_F^2}\,\mathrm{d}E = \frac{g_s g_V}{2\pi\hbar^2 v_F^2}\frac{1}{2}E_F^2 = \frac{E_f^2}{\pi\hbar^2 v_F^2}$$

where $E_F = \pm\hbar v_F\sqrt{|n|\pi} = \pm\hbar v_F k_F$.

11.6 Massless Dirac fermions

$$m^* = \frac{1}{\hbar}/\frac{\partial^2 E}{\partial k^2}$$

For $E = \hbar v_F k$, $\frac{\partial E}{\partial k} = \hbar v_F$, and $\frac{\partial^2 E}{\partial k^2} = 0$, the effective mass will diverge. So, for massless particle, $m_0 = 0$, $E = c|p|$.

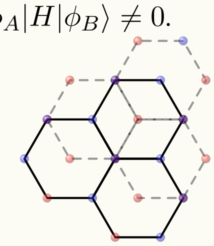
11.7 Bilayer Graphene

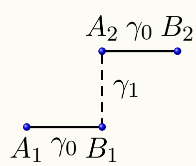
The matrix formations of Schrödinger equation

$$\begin{pmatrix} H_{AA} & H_{AB} \\ H_{BA} & H_{BB} \end{pmatrix} \psi = \begin{pmatrix} \langle\phi_A|\phi_A\rangle & \langle\phi_A|\phi_B\rangle \\ \langle\phi_B|\phi_A\rangle & \langle\phi_B|\phi_B\rangle \end{pmatrix} \psi \tag{11.13}$$

For the nearest hopping, the vectors are defined: $\boldsymbol{\delta}_1$, $\boldsymbol{\delta}_2$, $\boldsymbol{\delta}_3$ with $\langle\phi_A|H|\phi_B\rangle \neq 0$.
There are two ways for stacking: $A-A$ stacking and $A-B$ stacking.
The former one is unstable, so we consider the latter one only.
Denote the hopping parameter between A_i and B_i is γ_0, so

$$\gamma_0 = \langle\phi_{A_1}|H|\phi_{B_1}\rangle = \langle\phi_{A_2}|H|\phi_{B_2}\rangle$$





also consider the hopping between A_i and B_j, another nearest hopping the hopping between A_i and A_j can be ignored. We can write the Schödinger with a labeled Hamiltonian matrix, the crossing elemets correspoondings to the hopping between the labels.

$$\begin{array}{cc} \begin{array}{cccc} A_1 & B_1 & A_2 & B_2 \end{array} & \\ \begin{pmatrix} \epsilon_0 - E & -\gamma_0 f(k) & 0 & 0 \\ -\gamma_0 f^*(k) & \epsilon_0 - E & \gamma_1 & 0 \\ 0 & \gamma_1 & \epsilon_0 - E & -\gamma_0 f(k) \\ 0 & 0 & -\gamma_0 f^*(k) & \epsilon_0 - E \end{pmatrix} & \begin{array}{c} A_1 \\ B_1 \\ A_2 \\ B_1 \end{array} \end{array} \boldsymbol{\psi} = 0$$

where ϵ_0 is a constant value: the site energy; and $f(k) = \sum_{i=1}^{3} \mathrm{e}^{-\mathrm{i}k\cdot\delta_k}$. We can obtain two energies

$$E_{\pm}^{\alpha} = \pm\frac{\gamma_1}{2}\left(\sqrt{1+\frac{4\gamma_0^2|f(k)|^2}{\gamma_1^2}}+\alpha\right) \tag{11.14}$$

where $\alpha = \pm 1$.

(a) $\alpha = +1$: k, k' valley, $f(k) = 0$,

$$E_{\pm}^{+1} = \pm\frac{\gamma_1}{2}\times 2 = \pm\gamma_1 \tag{11.15}$$

(b) $\alpha = -1$: $f(k) = 0$

$$E_{\pm}^{-1} = \pm\frac{\gamma_1}{2}(1-1) = 0, \tag{11.16}$$

and the two bands will touch each other.

Around K point, $f(\boldsymbol{k}) \to 0$, so $|\gamma_0 f(\boldsymbol{k})| \ll |\gamma_1|$. Denoting $m^* = \frac{\gamma_1}{2v_F^2}$, so, the 2×2 Hamiltonian can be expressed by

$$H_2 = \begin{pmatrix} 0 & -\frac{\gamma_0^2 f^2(\boldsymbol{k})}{\gamma_1} \\ -\frac{\gamma_0^2(f^*(\boldsymbol{k}))}{\gamma_1} & 0 \end{pmatrix} = -\frac{1}{2m^*}\begin{pmatrix} 0 & (\xi p_x - \mathrm{i}p_y)^2 \\ (\xi p_x + \mathrm{i}p_y)^2 & 0 \end{pmatrix} = -\frac{p^2}{2m^2}\boldsymbol{\sigma}\cdot\boldsymbol{n}_2 \tag{11.17}$$

One can also call m^* the massive Dirac fermions.

Extension For N-layer graphene: $E(k) \sim k^N$. The energy bands get flatter as N increasing, which will get a larger density of states.

11.7.1 Bilayer graphere under an electric field

By applying an electric field parallels to z-axis, the on-site energies will be different

$$\langle\phi_{A_1}|H|\phi_{A_1}\rangle \neq \langle\phi_{A_2}|H|\phi_{A_2}\rangle$$

There will a band gap Δ.

11.8 Landau Levels in Graphene

Previous, we've discussed the free electron, so, the Landau quantization

$$E = \hbar\omega\left(N+\frac{1}{2}\right)$$

which is based on the parabolic relation $\frac{p^2}{2m}$.
In mono graphene, since the Hamiltonian

$$H = v_F\boldsymbol{\sigma}\cdot\boldsymbol{p}$$

we will get a different dispersion relation. Starting from

$$H = v_F\boldsymbol{\sigma}\cdot\boldsymbol{p} = v_F\begin{pmatrix} 0 & p_x - \mathrm{i}p_y \\ p_x + \mathrm{i}p_y & 0 \end{pmatrix} \tag{11.18}$$

Replace $\boldsymbol{p} \to -\boldsymbol{A}$, then using Landau gauge: $A_x = 0$, $A_y = Bx$, $A_z = 0$

$$q_x \to z_x,\ q_y \to q_y - x/l_B^2,\ l_B = \sqrt{\hbar/eB}$$

The Hamiltonian becomes

$$H = \hbar v_F \begin{pmatrix} 0 & q_x - \mathrm{i}\left(q_y - \frac{x}{l_B^2}\right) \\ q_x + \mathrm{i}\left(q_y - \frac{x}{l_B^2}\right) & 0 \end{pmatrix}$$

Define the operator $a = \frac{1}{\sqrt{2} l_B}(\tilde{x} + \mathrm{i} l_B^2 q_x)$, we have $[a, a^\dagger] = 1$. The Hamiltonian

$$H = \hbar v_F \begin{pmatrix} 0 & q_x + \frac{\mathrm{i}\tilde{x}}{l_B^2} \\ q_x - \frac{\mathrm{i}\tilde{x}}{l_B^2} & 0 \end{pmatrix} = \frac{\sqrt{2}\hbar v_F}{l_B} \begin{pmatrix} 0 & \mathrm{i} a^\dagger \\ -\mathrm{i} a & 0 \end{pmatrix}$$

from 1D quantm harmonic oscillator, we *guess* the eigenstates to be

$$|\psi_n\rangle = \begin{pmatrix} A_n|n\rangle \\ B_n|n-1\rangle \end{pmatrix}$$

Apply H to it, one can obtain

$$B_n \sqrt{2}\hbar v_F\, \mathrm{i}\sqrt{n} = E A_n,$$
$$A_n \frac{\sqrt{2}\hbar v_F}{l_B}(-\mathrm{i}\sqrt{n}) = E B_n.$$

Multiply them, we can obtain the energy

$$E_n = \pm v_F \sqrt{2\hbar e B|n|}$$

which is promotional to $\sqrt{B}$.

Lecture #1 Free electron model

Problem 1.1. Two-charge-carrier Drude model:
Consider a system of two types of charge carriers in the Drude model. The two carriers have the same density (n) and opposite charge (e and $-e$), and their massesand relaxation rates are m_1, m_2 and τ_1, τ_2, respectively. So their corresponding mobilities are $\mu_1 = \frac{e\tau_1}{m_1}$ and $\mu_2 = \frac{e\tau_2}{m_2}$, respectively.

(a) Calculate the magnetoresistance $\Delta\rho = \rho(B) - \rho(B=0)$, where B is the magnetic field.

(b) Calculate the Hall coefficient.

(c) In an undoped semiconductor, $n = n_0\, \mathrm{e}^{-\Delta/k_B T}$ describes the temperature dependence of the carrier concentration. What will the temperature dependence of the magnetoresistance and the Hall coefficient be?

Solution (By TA)**.** The EOM

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = q(\boldsymbol{E} + \boldsymbol{v} \times B) - m\frac{v}{\tau} = 0$$

Apply $\boldsymbol{j} = \hat{\sigma}\boldsymbol{E} = nq\boldsymbol{v}$, then

$$\boldsymbol{E} = \frac{m}{mq^2\tau}\boldsymbol{j} - \frac{1}{qn}\boldsymbol{j} \times \boldsymbol{B} = \begin{pmatrix} \frac{m}{nq^2\tau} & -\frac{B}{qn} & 0 \\ \frac{B}{qn} & \frac{m}{nq^2\tau} & 0 \\ 0 & 0 & \cdots \end{pmatrix}$$

The conductivity

$$\hat{\sigma} = \boldsymbol{E}^{-1} = \begin{pmatrix} \frac{\sigma_0}{1+\omega_c^2\tau^2} & \sigma_0\frac{\omega_c\tau}{1+\omega_c^2\tau} & 0 \\ -\sigma_0\frac{\omega_c\tau}{1+\omega_c^2\tau^2} & \frac{\sigma_0}{1+\omega_c^2\tau^2} & \end{pmatrix}$$

where $\omega_c = \frac{Bq}{m}$. When $B = 0$,

$$\rho_{xx}(0) = \frac{1}{\sigma_{10} + \sigma_{20}}, \quad \rho_{xy}(0) = 0$$

When $B \neq 0$

$$\rho_{xx} = \frac{\sigma_{xx}}{\sigma_{xx}^2 + \sigma_{xy}^2}, \quad \rho_{xy} = -\frac{\sigma_{xy}}{\sigma_{xx}^2 + \sigma_{xy}^2}$$

Solution.

(a) For the two carriers, we have

$$\sigma_{0i} = \frac{ne^2\tau_i}{m_i} = ne\mu_i, \quad \omega_{ci} = \frac{eB}{m_i}, \quad \omega_{ci}\tau_i = \mu_i B$$

The total conductivity tensor is

$$\hat{\sigma} = \hat{\sigma}_1 + \hat{\sigma}_2$$

and the components are

$$\sigma_{xx} = \frac{ne\mu_1}{1+(\mu_1 B)^2} + \frac{ne\mu_2}{1+(\mu_2 B)^2}$$
$$\sigma_{xy} = \frac{ne\mu_1^2 B}{1+(\mu_1 B)^2} + \frac{ne\mu_2^2 B}{1+(\mu_2 B)^2}$$
$$\sigma_{zz} = ne(\mu_1 + \mu_2)$$

The resistance tensor is the inverse

$$\hat{\rho} = \hat{\sigma}^{-1} = \frac{1}{\sigma_{xx}^2 + \sigma_{xy}^2} \begin{pmatrix} \sigma_{xx} & -\sigma_{xy} & 0 \\ \sigma_{xy} & \sigma_{xx} & 0 \\ 0 & 0 & \frac{\sigma_{xx}^2+\sigma_{xy}^2}{\sigma_{zz}} \end{pmatrix}.$$

Then, the element ρ_{xx} becomesrt

$$\rho_{xx}(0) = \sigma_{xx}^{-1}(0) = \frac{1}{ne(\mu_1+\mu_2)}, \quad \text{and} \quad \rho_{xx} = \frac{\sigma_{xx}(B)}{\sigma_{xx}^2(B) + \sigma_{xy}^2(B)}$$

So, the magnetoresistance is

$$\Delta\rho = \rho_{xx}(B) - \rho_{xx}(0) = \frac{1}{ne}\left[\frac{\frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2}}{\left(\frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2}\right)^2 + B^2\left(\frac{\mu_1}{1+(\mu_1 B)^2} - \frac{\mu_2}{1+(\mu_2 B)^2}\right)^2} - \frac{1}{\mu_1+\mu_2}\right]$$

(b) Since

$$\rho_{xy}(B) = -\frac{\sigma_{xy}(B)}{\sigma_{xx}(B)^2 + \sigma_{xy}(B)^2},$$

subsitute it into the definition of Hall coefficient $R_H = \frac{\rho_{xy}}{B}$, that is

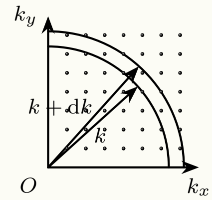
$$R_H = -\frac{1}{ne} \cdot \frac{\frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2}}{\left(\frac{\mu_1}{1+(\mu_1 B)^2} + \frac{\mu_2}{1+(\mu_2 B)^2}\right)^2 + B^2\left(\frac{\mu_1}{1+(\mu_1 B)^2} - \frac{\mu_2}{1+(\mu_2 B)^2}\right)^2}$$

(c) For simplicity, consider the case B is small. Now, $\Delta\rho$ and R_H becomes

$$\Delta\rho = ne\mu_1\mu_2(\mu_1 + \mu_2)\left(\frac{\mu_1 - \mu_2}{\mu_1 + \mu_2}\right)^2 B^2, \quad \text{and} \quad R_H = -\frac{1}{ne}\frac{\mu_1 - \mu_2}{\mu_1 + \mu_2}$$

So, in this case, the magnetoresistance $\Delta\rho \propto \mathrm{e}^{-\Delta/k_B T}$, while the Hall coefficient $R_H \propto \mathrm{e}^{\Delta/k_B T}$.

Problem 1.2. Density of States in 2D system:
Consider free electron in a 2D box with a width of a. The allowed wave-numbers k_x and k_y for x and y directions respectively are expressed as $\frac{\pi n_x}{a}$ and $\frac{\pi n_y}{a}$, where $n_x, n_y = 1, 2, 3, \ldots$ are integer numbers. The following figure shows the schematic view of the 2D array of allowed quantum states in k-space.



(a) Determine the area in k-space occupied by one allowed k point.

(b) Express number of allowed states in the area enclosed by the first quarter circles with the radus of k and $k + \mathrm{d}k$, where $k \gg \mathrm{d}k$ (Consider the spin degeneracy).

(c) Express density of allowed states in real space per unit energy.

Solution.

(a) The spaces of unit k point in k_x-direction and k_y-direction are

$$\Delta k_x = \frac{\pi}{a}, \quad \text{and} \quad \Delta k_y = \frac{\pi}{a}$$

Therefore, the area in k-space occupied by one allowed k point is

$$\Delta k_x \cdot \Delta k_y = \left(\frac{\pi}{a}\right)^2$$

(b) The area enclosed by the two quarter circles is

$$\mathrm{d}A = \frac{1}{4} 2\pi k \, \mathrm{d}k = \frac{\pi k}{2} \, \mathrm{d}k$$

Since every k point occupy an area of $(\pi/a)^2$, so the number of allowed k-points in this area is

$$\mathrm{d}N = \frac{2 \, \mathrm{d}A}{(\pi/a)^2} = \frac{ka^2}{\pi} \, \mathrm{d}k$$

Since two electrons can occupy each grid point ($|\uparrow\rangle$ and $|\downarrow\rangle$).

(c) Derivate it to the energy for a free election, we get

$$\mathrm{d}E = \mathrm{d}\left(\frac{\hbar^2 k^2}{2m}\right) = \frac{\hbar^2 k}{m} \, \mathrm{d}k, \quad \mathrm{d}k = \frac{m \, \mathrm{d}E}{\hbar^2 k^2}.$$

Substitute it into the result in (b): the number of states in $\mathrm{d}k$

$$\mathrm{d}N = \frac{ka^2}{\pi} \, \mathrm{d}k = \frac{ma^2}{\pi \hbar^2} \, \mathrm{d}E,$$

means in the whole system of area a^2, a number $\mathrm{d}N$ of states are allowed in energy range $\mathrm{d}E$. So, the density of allowed states in real space per unit energy is

$$g(E) = \frac{1}{a^2} \frac{\mathrm{d}N}{\mathrm{d}E} = \frac{m}{\pi \hbar^2}.$$

Problem 1.3. Density of states from a general formula:

(a) In a 3D solid and starting from the general formula, $g(E) = \frac{2V}{(2\pi)^3} \iint \frac{\mathrm{d}S_E}{\nabla_k E}$, find the expression for the electronic density of states: (i) for free electrons; (ii) for electrons that obey the isotropic dispersion relation: $E = -E_0 - 2\gamma \cos\left(\frac{ka}{2}\right)$, where $E_0, \gamma > 0$.

(b) After having established the corresponding expression for 2D, find $g(E)$ for (i) and (ii) above.

(c) Same question for 1D.

Solution (By TA).

$$\mathrm{d}V_E = \mathrm{d}S_E \cdot \mathrm{d}k = \mathrm{d}S_E \cdot \frac{\mathrm{d}E}{|\nabla_k E|}$$

Then, the volume of the sphere surface

$$V_{\text{site}} = \mathrm{d}E \iint \frac{\mathrm{d}S}{|\nabla_k E|}$$

Also for

$$\mathrm{d}N = 2\frac{1}{\left(\frac{2\pi}{a}\right)^3}, \quad \mathrm{d}N = g(E)\,\mathrm{d}E$$

For 2D case

$$g(E) = 2\frac{A}{(2\pi)^2}\int \frac{\mathrm{d}L}{|\nabla_k E|}$$

1D

$$g(E) = 2 \cdot \frac{L}{2\pi}\frac{1}{|\nabla_k E|}$$

Solution. Firstly, we shall calculate something before: For the isotropic dispersion

$$E = -E_0 - 2\gamma\cos(ka/2),$$

we can obtain

$$\sin\left(\frac{ka}{2}\right) = \sqrt{1 - \left(\frac{E+E_0}{2\gamma}\right)^2}, \quad \text{and} \quad k = \frac{2}{a}\arccos\left(-\frac{E+E_0}{2\gamma}\right).$$

Since $E = \hbar^2 k^2/2m$, so $k = \sqrt{2mE/\hbar^2}$, $\nabla_k E = \hbar^2 k/m$, $\nabla_k E = \gamma a \sin(ka/2)$ for isotropic dispersion.

(a) In a 3D situation, the general formula becomes

$$g(E) = \frac{2V}{(2\pi)^3} \cdot \frac{4\pi k^2}{\hbar^2 k/m} = \frac{Vmk}{\pi^2\hbar^2} = \frac{Vm}{\pi^2\hbar^3}\sqrt{2mE}.$$

For isotropic dispersion, the general formula becomes

$$g(E) = \frac{Vk^2}{\pi^2\gamma a\sqrt{1-(\frac{E+E_0}{2\gamma})^2}}.$$

(b) In a 2D situation, the general formula becomes

$$g(E) = \frac{2A}{(2\pi)^2} \cdot \frac{2\pi k}{\hbar^2 k/m} = \frac{mA}{\pi\hbar^2} = \text{Const.}$$

For isotropic dispersion, the general formula becomes

$$g(E) = \frac{2A}{(2\pi)^2} \cdot \frac{2\pi k}{\gamma a \sin(ka/2)} = \frac{2A}{\pi\gamma a^2} \frac{\arccos\left(-\frac{E+E_0}{2\gamma}\right)}{\sqrt{1-\left(\frac{E+E_0}{2\gamma}\right)^2}}$$

(c) In 1D situation, $g(E) = \frac{2L}{2\pi} \cdot \frac{2}{\mathrm{d}E/\mathrm{d}k}$. The general formula becomes

$$g(E) = \frac{2L}{2\pi} \cdot \frac{2}{\mathrm{d}E/\mathrm{d}k} = \frac{2L}{\pi \ \mathrm{d}E/\mathrm{d}k} = \frac{L}{\pi\hbar}\sqrt{\frac{2m}{E}}$$

For isotropic dispersion, the general formula becomes

$$g(E) = \frac{2L}{\pi\gamma a} \frac{1}{\sqrt{1-\left(\frac{E+E_0}{2\gamma}\right)^2}}.$$

Lecture #2 Phonon

Problem 2.1 (Normal modes of a square lattice)**.** Consider a square lattice of atoms of mass M, in which the nearest neighbors are coupled to each other by springs with constants K. Derive the equation of motion for the mode in which atomic displacements are normal to the crystal plane and find the dispersion of this mode. Analyze the behavior of this mode in the continuum limit.

Solution (By TA)**.** Only consider the vibration perpendicular to the surface

$$M\frac{\mathrm{d}^2U_{n,m}}{\mathrm{d}t^2} = -k\sum[u_{n,m} - u_{n\pm1,m\pm1}]$$

Substitute the plane wave

$$u_{n,m}(t) = U\,\mathrm{e}^{\mathrm{i}(q_x na + q_y na - \omega t)}$$

we have

$$-M\omega^2 = 2k[\cos(q_x a) + \cos(q_y a) - 2]$$

For $|q_x a|, |q_y a| \ll q, \sin qa = qa$.

Solution. Assume the lattice constant a, then the lattice sites become

$$\boldsymbol{R}_{n,m} = (na, ma), \quad n, m \in \mathbb{Z}.$$

We consider every atom has a single degree of freedom: the out-of-plane scalar displacement $u_{n,m}(t)$. So, the spring connecting site (n, m) and a neighbour (n', m') has extension $\Delta = u_{n',m'} - u_{n,m}$, which brings the force to atom (n, m)

$$F_{n,m} = -K(u_{n,m} - u_{n',m'})$$

due to Hooke's law. Since every atom has four nearest neighbours: $(n \pm 1, m)$, $(n, m \pm 1)$, which bring four forces to the "central atom". Summing these forces

$$F^{\text{tot}}_{n,m} = -\sum_{\langle n,m;n',m'\rangle} K(u_{n,m} - u_{n',m'}) = -K(4u_{n,m} - u_{n+1,m} - u_{n-1,m} - u_{n,m+1} - u_{n,m-1})$$

According to Newton's second law, we obtain the EOM

$$M\ddot{u}_{n,m} = -K(4u_{n,m} - u_{n+1,m} - u_{n-1,m} - u_{n,m+1} - u_{n,m-1}).$$

Assume the EOM has a plane-wave solution

$$u_{n,m}(t) = A\,\mathrm{e}^{\mathrm{i}(k_x na + k_y ma - \omega t)},$$

then substitute it into the EOM, the LHS is

$$\text{LHS} = M(-\omega^2)u_{n,m}$$

and the RHS is

$$\text{RHS} = -K(4 - \mathrm{e}^{\mathrm{i}k_x a} - \mathrm{e}^{-\mathrm{i}k_x a} - \mathrm{e}^{\mathrm{i}k_y a} - \mathrm{e}^{-\mathrm{i}k_y a})u_{n,m} = -2K[2 - \cos(k_x a) - \cos(k_y a)]u_{n,m}$$

Finally we have the dispersion of this model

$$\omega^2(\boldsymbol{k}) = \frac{2K}{M}[2 - \cos(k_x a) - \cos(k_y a)], \quad \omega(\boldsymbol{k}) = 2\sqrt{\frac{K}{M}}\sqrt{\sin^2\left(\frac{k_x a}{2}\right) + \sin^2\left(\frac{k_y a}{2}\right)}.$$

In the continuum limit, i.e., $|\boldsymbol{k}|a \ll 1$, then the dispersion of this model becomes linear to $|\boldsymbol{k}|$

$$\omega(\boldsymbol{k}) \approxeq 2\sqrt{\frac{K}{M}}\sqrt{\left(\frac{k_x a}{2}\right)^2 + \left(\frac{k_y a}{2}\right)^2} = c|\boldsymbol{k}|$$

where $c = a\sqrt{K/M}$.

Problem 2.2 (Localized phonons on an impurity). Consider a linear chain of identical atoms of mass m each equidistant by a.

(a) Limiting the problem to interactions between nearest neighbors characterized by the force constant β, give the dispersion relation of longitudinal phonons and derive the expression of their group velocity. Express the results as a function of "a" and of the velocity of sound, v_s. What is maximum frequency of atomic vibration, ω_M, when $v_s = 5000\,\mathrm{m/s}$?

(b) An atom in the middle of the chain is replaced by the atom of an impurity with a mass m' (where $m' < m$). Write the displacement equations, u_0, for this impurity atom and the displacement "u_n" for an atom located at a distance "na". Assume that the force constant β between nearesst neighbors is the same regardless of the atom considered.

(c) The impurity atom being lighter than the others has an oscillation frequency ω'_m larger than ω_M and it drags the other neighbor atoms by its motion. We try to describe its effect with the expression of a damped wave of form: $u_n = u_0\,\mathrm{e}^{-\alpha|x|}\,\mathrm{e}^{\mathrm{i}(\omega' t - kx)}$. What is the value of the frequency ω'_M, which is compatible with the equations of motion when the wave vector is near the boundary of the first Brillouin Zone? Verify that when $m = m'$, the solution matches that obtained in (a).

(d) After having numerically evaluated ω'_M and a when $m' = m/2$, sketch the instantaneous position of atoms as a function of x.

Solution (By TA). $v_y = \frac{\mathrm{d}m}{\mathrm{d}k}$. v_s is long-wave limit v_s: $\lim_{k\to 0} v_g = a\sqrt{\frac{\beta}{m}}$, and $\omega_m = \frac{2v_s}{a}$.

$$u_n = u_0\,\mathrm{e}^{-\alpha na}\,\mathrm{e}^{\mathrm{i}(\omega' t - kna)}$$

Substitute it into the EOM

$$\begin{cases} -\omega'^2 m' = \beta[\mathrm{e}^{ax}(\mathrm{e}^{\mathrm{i}ka} + \mathrm{e}^{-\mathrm{i}ka}) - 2] \\ -\omega'^2 m = \beta[\mathrm{e}^{(\alpha+\mathrm{i}k)a} + \mathrm{e}^{-(\alpha+\mathrm{i}k)a} - 2] \end{cases}$$

where $ka = \pi$, $\omega' = \omega'_m$,

$$u_n = u_0(-1)^n\,\mathrm{e}^{\mathrm{i}\omega t}\,\mathrm{e}^{-\alpha a|n|}$$

$$\begin{cases} m\omega'^2_m = 2\beta(1 + \mathrm{e}^{-\alpha a}), \\ m\omega' m^2 = \Delta\beta\,\mathrm{ch}^2\left(\frac{\alpha a}{2}\right) \end{cases}$$

Solution.

(a) Assume the nearest-neighbor spring constant β. The EOM for the n-th atom is

$$m\ddot{u}_n = -\beta(u_n - u_{n+1}) - \beta(u_n - u_{n-1}) = \beta(u_{n+1} - u_{n-1} - 2u_n).$$

We assume the EOM has a plane-wave solution $u_n(t) = A\,\mathrm{e}^{\mathrm{i}(kna-\omega t)}$, then substitute it into the EOM

$$-m\omega^2 u_n = -\beta(2 - \mathrm{e}^{\mathrm{i}ka} - \mathrm{e}^{-\mathrm{i}ka})u_n = -\beta(2 - 2\cos(ka))u_n = -4\beta\sin^2\left(\frac{ka}{2}\right),$$

we have the dispersion

$$\omega = 2\sqrt{\frac{\beta}{m}}\left|\sin\left(\frac{ka}{2}\right)\right|.$$

The group velocity and the velocity of sound are

$$v_g(k) = \frac{\mathrm{d}\omega}{\mathrm{d}k} = a\sqrt{\frac{\beta}{m}}\cos\left(\frac{ka}{2}\right), \quad \text{and} \quad v_s(a) = \lim_{k\to 0} v_g(k) = a\sqrt{\frac{\beta}{m}}$$

then the group velocity, as well as the frequency can be expressed as

$$v_g(k) = v_s\cos\left(\frac{ka}{2}\right), \quad \text{and} \quad \omega = \frac{2v_s}{a}\left|\sin\left(\frac{ka}{2}\right)\right|.$$

when $k = \pi/a$, the frequency arrives at a maximal value

$$\omega_M = \frac{2v_s}{a} \xlongequal{v_s=5000\,\mathrm{m/s}} \frac{10000}{a}\mathrm{s}^{-1}.$$

(b) The impurity is located at site $n = 0$, with mass m'; while its neighbors $n = \pm 1$ with mass m. The EOM for the impurity and other atoms are

$$\begin{aligned} m'\ddot{u}_0 &= -\beta(u_1 - u_0) + \beta(u_{-1} - u_0) = \beta(u_1 + u_{-1} - 2u_0) \\ m\ddot{u}_n &= \beta(u_{n+1} + u_{n-1} - 2u_n) \end{aligned}$$

(c) In (a), near the boundary of the first BZ, i.e. $k = \pi/a$, the frequency is $\omega_\mathrm{I} = 2\sqrt{\beta/m}$. Write the given wave form in terms of site number n

$$u_n = A\,\mathrm{e}^{-\alpha|n|a}\,\mathrm{e}^{\mathrm{i}(\omega' t - kna)}$$

Substitute it into the two EOMs in (b), and let $n = 1$

$$\begin{aligned} -\omega^2 m' u_0 &= \beta(\mathrm{e}^{-\alpha a - \mathrm{i}ka} + \mathrm{e}^{-\alpha a + \mathrm{i}ka} - 2)u_0 \xlongequal{k=\pi/a} -2\beta(\mathrm{e}^{-\alpha a} + 1)u_0 \\ -\omega^2 m u_1 &= \beta(u_2 + u_0 - 2u_1) = \beta(\mathrm{e}^{-\alpha a - \mathrm{i}ka} + \mathrm{e}^{\alpha a + \mathrm{i}ka} - 2)u_1 \xlongequal{k=\pi/a} -2\beta(\mathrm{ch}(\alpha a) + 1)u_1 \end{aligned}$$

Derive the two expressions, we have

$$\frac{m'}{m} = \frac{\mathrm{e}^{-\alpha a} + 1}{\mathrm{ch}(\alpha a) + 1} = \frac{2}{\mathrm{e}^{\alpha a} + 1}, \quad \text{and} \quad \mathrm{e}^{-\alpha a} + 1 = \frac{2m}{2m - m'}$$

Substitute it into one of the EOM, we obtain the dispersion

$$\omega_M'^2 = \omega'^2 = \frac{4\beta m}{m'(2m - m')}$$

where $\omega_M'^2 = \omega'^2$ is due to the boundary of the first BZ ($k = \pi/a$). When $m' = m$, the frequency becomes

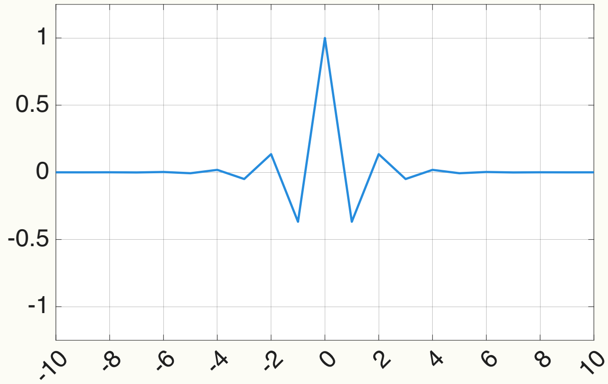
$$\omega_M'^2 = \frac{4\beta}{m}, \quad \text{and} \quad \omega_M' = 2\sqrt{\frac{\beta}{m}} = \omega_\mathrm{I}$$

which matchs that obtained in (a).

(d) Substitute $m' = m/2$ in the result of (c), we have $\omega_M'^2 = \frac{16}{3}\frac{\beta}{m}$. To simplify, we choose the time $t = 0$. Near the first BZ, the displacement of atoms is

$$u_n = (-1)^n A\, \mathrm{e}^{-\alpha|n|a}$$

The plotted figure of atoms is attached as follows (Just show the relative position for reference).



Problem 2.3 (Density of states and specific heat of a monoatomic 1D lattice). Consider a row of N identical atoms of mass m and equidistant by a.

(a) Limiting the problem to interactions between nearest neighbors with a force constant β, find the expression for the dispersion relation of acoustic longitudinal vibrations propagating along this row. What is the maximum frequency ω_M of atomic vibrations, written in terms of v_s and a? Compare the value obtained with that of ω_D deduced from the Debye model.

(b) Find the corresponding expression for the density of states $g(\omega)$.

(c) Write the expressions for internal energy U and for the heat capacity C_v. Find the limits as these functions approach high temperatures ($\hbar\omega_M \ll k_B T$).

(d) What is the evolution of C_v at low temperature? Compare the result obtained here with that deduced from the Debye model.

Solution (By TA). (a) $\omega = v_s k$, $\int_0^{\omega_0} g_D(\omega)\,\mathrm{d}\omega = N$, $\omega(k) = 2\sqrt{\beta/m}|\sin(ka/2)|$.

$$g_D(\omega) = \frac{L}{\pi}\frac{\mathrm{d}k}{\mathrm{d}\omega} = \frac{L}{\pi}\cdot\frac{1}{v_s} = \frac{Na}{\pi v_s}$$

$\omega_D = \pi v_s/a$, $\omega_M = 2v_s/a$.

(b) $\langle n\rangle = \frac{1}{\mathrm{e}^{\hbar\omega/k_BT}-1}$, $\langle E\rangle = \langle n\rangle\hbar\omega$, $u_0 = \frac{1}{2}\hbar\omega$.

$$U = \int_0^{\omega_M} g(\omega)\left[\frac{1}{2}\hbar\omega + \frac{\hbar\omega}{\mathrm{e}^{\hbar\omega/k_BT}-1}\right]\mathrm{d}\omega$$

$D_v = \frac{\mathrm{d}U}{\mathrm{d}T}$, for $\hbar\omega \ll k_BT$, the exponential is $\frac{\hbar\omega}{k_BT}$,

$$C_v = k_B\int_0^{\omega_a} g(\omega)\,\mathrm{d}\omega = Nk_B$$

(c) Only $\omega \to 0$ contributes to the integration.

$$\mathrm{e}^{\hbar\omega/k_BT}\begin{cases}\hbar\omega \gg k_BT\frac{\hbar\omega}{\mathrm{e}^{\hbar\omega k_BT}-1} = \hbar\omega\,\mathrm{e}^{-\hbar\omega/k_BT},\\ \hbar\omega \ll k_BT\,\mathrm{e}^{\hbar\omega/k_BT} \approx k_BT\end{cases}$$

$\omega \to 0$, $g(\omega) = g(0) = \frac{2N}{\pi\omega_M}$.

Solution.

(a) The same to Problem 2.2(a): the dispersion

$$\omega(k) = 2\sqrt{\frac{\beta}{m}}\left|\sin\left(\frac{ka}{2}\right)\right|,\ v_s = \lim_{k\to 0}\frac{\mathrm{d}\omega}{\mathrm{d}k} = a\sqrt{\frac{\beta}{m}},\quad \text{and}\quad \omega_M \xlongequal{k=\pi/a} 2\sqrt{\frac{\beta}{m}} = \frac{2v_s}{a},$$

and the Debye frequency for 1D chain is $\omega_D = v_s k = \frac{\pi v_s}{a}$. Since $\omega_M/\omega_D = 2/\pi < 1$, the maximum frequency in the real dispersion is smaller than in the Debye model.

(b) In 1D chain, the number of states within $\mathrm{d}k$ is $2 \cdot \frac{L}{2\pi}\,\mathrm{d}k$. So, the density of states is

$$g(\omega) = \frac{L}{\pi}\frac{\mathrm{d}k}{\mathrm{d}\omega} = \frac{2L}{\pi a \omega_M}\frac{1}{\sqrt{1-(\omega/\omega_M)^2}} \xlongequal{L=Na} \frac{2N}{\pi\omega_M}\frac{1}{\sqrt{1-(\omega/\omega_M)^2}}.$$

(c) The phonon thermal occupation

$$n(\omega) = \left[\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1\right]^{-1}$$

Then the internal energy is

$$U(T) = \int_0^{\omega_M} \hbar n(\omega) g(\omega)\,\mathrm{d}\omega = \int_0^{\omega_M} \frac{\hbar\omega}{\mathrm{e}^{\frac{\hbar\omega}{k_B T}} - 1} g(\omega)\,\mathrm{d}\omega.$$

and the heat capacity

$$C_V(T) = \frac{\mathrm{d}U}{\mathrm{d}T} = \int_0^{\omega_M} \frac{\partial n(\omega)}{\partial T} g(\omega)\,\mathrm{d}\omega$$

At high-temperature limit, i.e., $\hbar\omega_M \ll k_B T$. The expression of internal energy and heat capacity become

$$U(T) = \int_0^{\omega_M} k_B T g(\omega)\,\mathrm{d}\omega = N k_B T,\ C_V(T) = N k_B.$$

where applied the total number of vibration modes $\int_0^{\omega_M} g(\omega)\,\mathrm{d}\omega = N$.

(d) At low temperature limit, only small k matters, i.e., the main contributions come from small ω. Then, $g(\omega) = \frac{L}{\pi v_s}$. The internal energy becomes

$$U = \frac{L}{\pi v_s} \int_0^\infty \frac{\hbar\omega}{\mathrm{e}^{\frac{\hbar\omega}{k_B T}} - 1} \, \mathrm{d}\omega = \frac{\pi L (k_B T)^2}{6\hbar v_s}$$

and the capacity

$$C_v = \frac{\mathrm{d}U}{\mathrm{d}T} = \frac{\pi N a k_B^2 T}{3\hbar v_s} \propto T$$

the same conclusion as the Debye model.

Lecture #3 Crystal lattice

Problem 3.1 (Crystallographic restriction theorem)**.** Proof the crystallographic restriction theorem: Consider a crystal (i.e. with translational symmetry) in three dimensions that is invariant under rotations by an angle $2\pi/n$ around an axis. Then the only allowed possibilities are $n = 1, 2, 3, 4, 6$ with the case $n = 1$ being the trivial case with no rotational symmetry.

Solution (By TA)**.** Rotation matrix

$$\mathbf{R} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The vector

$$A = (\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3)$$

after rotation

$$(L_1, L_2, L_3)^\mathsf{T} = A^{-1}RA(L_1, L_2, L_3)^\mathsf{T}$$

Since $\mathrm{Tr}(A^{-1}RA) = \mathrm{Tr}(R)$, then, $2\cos\alpha + 1 = \mathbb{Z}$.

Solution. *Proof.* Denote the 3D lattice can be generated by the basis vectors $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, and $\boldsymbol{e}_3$. So, any lattice point vector set should be

$$\boldsymbol{L} = \{a_1\boldsymbol{e}_1 + a_2\boldsymbol{e}_2 + a_3\boldsymbol{e}_3\}.$$

Obviously, $a_i \in \mathbb{Z}$, i.e., any lattice point can be labeled by integer coordinates (a_1, a_2, a_3). Consider the 3D rotation matrix (operator)

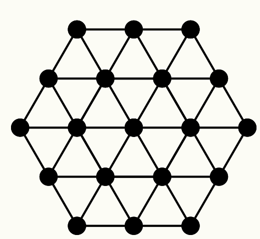
$$\mathbf{R} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

When acting it on the vector $\boldsymbol{L}$, it should leave the lattice invariant, i.e., the trace must be integers

$$\mathrm{Tr}(\mathbf{R}) = 2\cos\theta + 1 \in \mathbb{Z}.$$

So we have $2\cos\theta = 2\cos(2\pi/n) \in \mathbb{Z}$, i.e., $\cos(2\pi/n) = -1, -1/2, 0, 1/2, 1$, then $n \in \{1, 2, 3, 4, 6\}$. □

Problem 3.2 (2D triangular lattice)**.** Let us consider a two-dimensional triangular lattice as shown in the following figure.



(a) Identify the primitive unit cell of the two-dimensional triangular lattice. Find the basis vectors.

(b) Construct the basis vectors of the reciprocal unit cell.

Solution (By TA). $\boldsymbol{a}_i \boldsymbol{b}_j = 2\pi\delta_{ij}$, $b_i = (b_{1x}, b_{1y})$.

$$(\boldsymbol{a}_1, \boldsymbol{a}_2)(\boldsymbol{b}_1, \boldsymbol{b}_2)^{\mathsf{T}} = 2\pi \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Solution.

(a) Denote the lattice constant a. Then the simplest choice of primitive lattice vectors is

$$\boldsymbol{a}_1 = a(1,0), \quad \boldsymbol{a}_2 = a\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right).$$

(b) Denote the two reciprocal-lattice primitive vectors $\boldsymbol{b}_1 = (b_{1_x}, b_{1_y})$, $\boldsymbol{b}_2 = (b_{2_x}, b_{1_2})$. Using the relation

$$\boldsymbol{a}_i \cdot \boldsymbol{b}_j = 2\pi\delta_{ij}.$$

To expand it

$$\boldsymbol{a}_1 \cdot \boldsymbol{b}_1 = 2\pi, \quad \boldsymbol{a}_2 \cdot \boldsymbol{b}_1 = 0, \quad \boldsymbol{a}_1 \cdot \boldsymbol{b}_2 = 2\pi, \quad \boldsymbol{a}_2 \cdot \boldsymbol{b}_2 = 2\pi,$$

we obtain the two reciprocal-lattice primitive vectors

$$\boldsymbol{b}_1 = \frac{2\pi}{a}\left(1, -\frac{1}{\sqrt{3}}\right), \quad \boldsymbol{b}_2 = \frac{2\pi}{a}\left(0, \frac{2}{\sqrt{3}}\right).$$

Problem 3.3 (Properties of the reciprocal lattice).

(a) Show that all reciprocal lattice vectors of the form $\boldsymbol{G} = h\boldsymbol{A} + k\boldsymbol{B} + l\boldsymbol{C}$ are perpendicular to the planes of the same indices (h,k,l) in real space.

(b) Show that the distance $d_{h,k,l}$ between two consecutive planes (h,k,l) is inversely proportional to $|\boldsymbol{G}_{h,k,l}|$.

(c) Find $d_{h,k,l}$ for a simple cubic lattice and an orthorhombic lattice ($a \neq b \neq c$, $\alpha = \beta = \gamma = \pi/2$).

Solution. ‘’

(a) *Proof.* Substitute the expressions of the reciprocal lattice vectors to $\boldsymbol{G}$

$$\boldsymbol{G} = \frac{2\pi}{V}[h(\boldsymbol{a}_2 \times \boldsymbol{a}_3) + k(\boldsymbol{a}_3 \times \boldsymbol{a}_1) + l(\boldsymbol{a}_1 \times \boldsymbol{a}_2)].$$

Since the Miller indices (h,k,l) correspond to a plane that intercepts the crystal axies at $\boldsymbol{a}_1/h$, $\boldsymbol{a}_2/k$, $\boldsymbol{a}_3/l$, then, any vector lying in the (h,k,l) plane can be written as a linear combination of two independent vectors in that plane, i.e.,

$$\boldsymbol{r}_1 = \frac{\boldsymbol{a}_2}{k} - \frac{\boldsymbol{a}_1}{h}, \quad \boldsymbol{r}_2 = \frac{\boldsymbol{a}_3}{l} - \frac{\boldsymbol{a}_2}{k}.$$

Then, the normal to this plane can be written as a cross product

$$\hat{\boldsymbol{n}}_{hkl} = \boldsymbol{r}_1 \times \boldsymbol{r}_2 = \frac{1}{kl}(\boldsymbol{a}_2 \times \boldsymbol{a}_3) + \frac{1}{hl}(\boldsymbol{a}_3 \times \boldsymbol{a}_1) + \frac{1}{hk}(\boldsymbol{a}_1 \times \boldsymbol{a}_2).$$

which is parallel to $\boldsymbol{G}$, implies that $\boldsymbol{G}$ is perpendicular to the (h,k,l) plane. □

(b) *Proof.* For the vector pointing to the $n+1$ and n-th (h,k,l) plane, it satisfies

$$\boldsymbol{G}\cdot\boldsymbol{r}_{n+1} = 2\pi(n+1), \quad \text{and} \quad \boldsymbol{G}\cdot\boldsymbol{r}_n = 2\pi n,$$

where $\boldsymbol{r}_{n+1} - \boldsymbol{r}_n = d\hat{\boldsymbol{n}}$. Combine the two relations

$$\boldsymbol{G}\cdot(\boldsymbol{r}_{n+1} - \boldsymbol{r}_n) = \boldsymbol{G}\cdot\hat{\boldsymbol{n}}d = 2\pi d.$$

So, we have

$$d = \frac{2\pi}{\boldsymbol{G}\cdot\hat{\boldsymbol{n}}} = \frac{2\pi}{|\boldsymbol{G}_{h,k,l}|},$$

i.e., the distance between two consecutive planes is inversely proportional to $|\boldsymbol{G}_{h,k,l}|$. □

(c) For a simple cubic lattice

$$\boldsymbol{a} = (a,0,0), \quad \boldsymbol{b} = (0,b,0), \quad \boldsymbol{c} = (0,0,c).$$

Then, the reciprocal vectors can be expressed as

$$\boldsymbol{G} = \frac{2\pi h}{a}\hat{\boldsymbol{x}} + \frac{2\pi k}{b}\hat{\boldsymbol{y}} + \frac{2\pi l}{c}\hat{\boldsymbol{z}},$$

and its magnitude

$$|\boldsymbol{G}| = 2\pi\sqrt{\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}}.$$

So, we obtain the distance $d = \frac{2\pi}{|\boldsymbol{G}|} = \left(\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2}\right)^{-1/2}$.

Problem 3.4 (Atomic planes and Miller indices: application to lithium)**.** The Bravais lattice of lithium is simple cubic with lattice parameter $a = 3.48\,\text{Å}$.

(a) Suppose that the atoms (assumed to be spheres) are placed along the $[111]$, represent the atomic distribution along the following faces (100), (110), (111), and (201).

(b) For each such 2D structure, state the direction and basis vector, $\boldsymbol{a}$ and $\boldsymbol{b}$ of the elementary lattice as well as the value of the angle, γ.

(c) Find the atomic concentration and the mass density of lithium ($A \approx 7$).

Solution.

(a) (b) The distribution, basis vector, and the angle of the faces are listed.

2D Structure	Distribution	$\boldsymbol{a}$	$\boldsymbol{b}$	γ
(100)	square	$(0,a,0)$	$(0,0,a)$	$90°$
(110)	rectangular	$(a,-a,0)$	$(0,0,a)$	$90°$
(111)	hexagonal	$(a,0,-a)$	$(0,a,-a)$	$60°$
(201)	rectangular	$(0,a,0)$	$(a,0,-2a)$	$90°$

(c) Since the atoms are placed along the $[111]$, then every cubic lattice's corners has an atom, i.e., it is equivalent that there is an atom in every lattice. So, the atomic concentration is

$$n = \frac{1}{V} = \frac{1}{a^3} = 2.3728 \times 10^{28}\,\mathrm{atoms/m^3}.$$

The mass density is

$$\rho = n m_{\mathsf{atom}} = \frac{nA}{N_A} = 275.816\,\mathrm{kg/m^3}.$$

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Lecture #4 X-ray diffraction

Problem 4.1 (Structure factor for a tri-atomic basis)**.** Consider a linear crystal with lattice constant a in which the basis consists of atom of species A at 0 and two atoms of species B at $1/3$ and $2/3$.

(a) Find the structure factor of such a crystal.

(b) X-rays of wavelength $\lambda(= a/4)$ irradiate this crystal at an oblique incidence ($45°$). Using the Ewald construction in the plane of incidence, show graphically, the diffraction directions and indicate the relative weights of the different diffracted intensities.

(c) State the physical significance of the vector $\boldsymbol{k}$ from which the extremity is on a plane passing through the origin of the reciprocal lattice.

Solution (By TA)**.** (a)

$$S_a = \sum_j f_j \,\mathrm{e}^{-\mathrm{i}\boldsymbol{a}\cdot\boldsymbol{r}_j} = f_A \,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot 0} + f_B(\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\frac{a}{3}\hat{x}} + \mathrm{e}^{-\mathrm{i}\boldsymbol{a}\cdot\frac{2}{3}a\hat{x}}) = f_A + 2f_B\cos(\frac{2\pi}{3}h) = \begin{cases} f_A + 2f_B, \\ f_A - f_B \end{cases}$$

(b) $k = \frac{2\pi}{\lambda} = \frac{8\pi}{a}$,

$$k\cos 45° - k\cos\theta = G_x$$

$\cos\theta = \frac{h+2\sqrt{2}}{4} \in [-1, 1]$.

Solution.

(a) Substitute $r_i = 0, \frac{a}{3}, \frac{2a}{3}$ into the definition of the structure factor

$$S(h) = f_A \,\mathrm{e}^{\mathrm{i}G\cdot 0} + f_B \,\mathrm{e}^{\mathrm{i}G\cdot\frac{a}{3}} + f_B \,\mathrm{e}^{\mathrm{i}G\cdot\frac{2a}{3}} \xlongequal{G=2\pi h/a} f_A + f_B\left(\mathrm{e}^{\mathrm{i}\frac{2\pi h}{3}} + \mathrm{e}^{\mathrm{i}\frac{4\pi h}{3}}\right)$$

$$\xlongequal{\text{Euler's formula}} \begin{cases} f_A + 2f_B, & h = 3n \\ f_A - f_B, & \text{otherwise} \end{cases}$$

(b) The wave number

$$k = \frac{2\pi}{\lambda} = \frac{8\pi}{a} = 4b_0,$$

where $b_0 = \frac{2\pi}{a}$ is the reciprocal lattice vector. The incident wavevector is

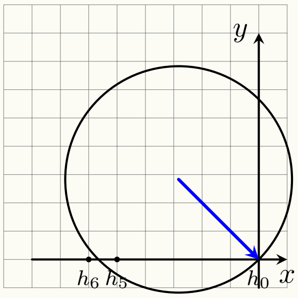
$$\boldsymbol{k}_i = \left(k/\sqrt{2}, -k/\sqrt{2}\right)$$

Then, the Ewald circle satisfies

$$\left(x + k/\sqrt{2}\right)^2 + \left(y - k/\sqrt{2}\right)^2 = k^2$$

and we can obtain the two intersection on x-axis: $x_1 = 0$ and $x_1 = -\sqrt{2}k$.

Obviously, $x_1 = 0b_0$, $n_1 = 0$; $x_2 = -5.656b_0$, $n_2 = -5.656$.



Since n is not an integer, no other reciprocal lattice points lies on the Ewald circle except the one on the original point. The intensity of each diffraction order is proportional to $|S(h)|^2$, i.e.,

$$I_h \propto |S(h)|^2 = \begin{cases} (f_A + 2f_B)^2, & h = 3m, \\ (f_A - f_B)^2, & \text{otherwise} \end{cases}$$

(c) The tip of $\boldsymbol{k}$ lying on a plane through the origin of the reciprocal lattice means the Bragg condition is satisfied for diffraction.

Problem 4.2. The main experimental methods of X-ray diffraction are

(a) the Laue method

(b) the crystal rotation method

(c) the powder method (or the Debye-Scherrer method)

Describe the corresponding experimental set-up and with the help of the Ewald construction in reciprocal space, deduce the observed diffraction patterns in each case.

Solution.

(a) Laue Method

i. Setup: Stationary single crystal, polychromatic X-rays.

ii. Ewald & Pattern: A continuous range of Ewald sphere radii (due to varying λ) means many reciprocal lattice points lie on some sphere. This produces a pattern of sharp spots, each from a different (hkl) plane and wavelength. The spot arrangement directly reveals the crystal's symmetry.

(b) Crystal Rotation Method

i. Setup: Rotating single crystal, monochromatic X-rays.

ii. Ewald & Pattern: The fixed Ewald sphere is intersected by rotating reciprocal lattice points. This generates discrete spots or arcs on the detector. Measuring the rotation angles and intensities of these spots allows for the determination of the unit cell and atomic structure.

(c) Powder Method

i. Setup: Powdered sample (random crystallites), monochromatic X-rays.

ii. Ewald & Pattern: Each set of (hkl) planes forms a spherical shell in reciprocal space. The Ewald sphere's intersection with these shells creates cones of diffracted beams (Debye cones), which appear as characteristic rings on the detector. The ring positions (2θ angles) provide d-spacings for phase identification and lattice parameter calculation.

Lecture #5 Nearly free electron model

Problem 5.1 (Compare Kronig-Penney model with the nearly free electron model)**.** The Kronig-Penney model allows one to consider both the nearly-free-electron and tight-binding limits. Consider the (repulsive) Dirac-Kronig-Penney model. It gives the following implicit equation for the energy bands

$$\cos(ka) = \cos(qa) + u\frac{\sin(qa)}{qa}$$

where $q = \sqrt{2mE}/\hbar$ and $u = \frac{mU_0 a}{\hbar^2} > 0$. Assuming that $u \ll q$, find the solution of above equation near the Bragg points $k \approx \pm\frac{\pi}{a}, \pm\frac{2\pi}{a}, \ldots$. Compare your result with that of the nearly-free electron model. (Reminder: the $\approx$ sign means that k is close yet no equal to the wavenumber of the Bragg point.)

Solution. KP model $k = \frac{n\pi}{a} + \delta$.

$$\cos ka = \cos qa + u\frac{\sin qa}{qa}$$

Since

$$\cos(n\pi + \delta a) = \cos n\pi \cos \delta a - \sin n\pi \sin \delta a = (-1)^n \cos \delta a = (-1)^n \left[1 - \frac{(\delta a)^2}{2}\right]$$

The energy

$$E^2 = \frac{\hbar^2}{2m}\left(\frac{n\pi}{a}\right)^2 + \Delta E, \quad q = \frac{\sqrt{2mE}}{\hbar} = \frac{n\pi}{a} + \epsilon$$

Then the cosine and sine

$$\cos(qa) = (-1)^n\left[1 - \frac{(\epsilon a)^2}{2}\right], \quad \frac{\sin(qa)}{qa} = \frac{(-1)^n \epsilon a}{n\pi}$$

We can obtain an equation about ϵ

$$\epsilon^2 - \frac{2U}{an\pi}\epsilon - \delta^2 = 0$$

NFE: Expand into a Fourier mode

$$V(x) = \sum_a V a\, \mathrm{e}^{\mathrm{i}Gx} = U_0 \delta(x - na)$$

$$k = \frac{n\pi}{a} + \Delta, \quad k' = \frac{n\pi}{a} - \Delta$$

The wavefunction

$$\psi(x) = a\psi_k^0 + b\psi_{k'}^0$$

where $\psi_k^0 = \frac{1}{\sqrt{L}}\,\mathrm{e}^{\mathrm{i}kx}$. The eigenequation

$$(H_0 + H')\psi(x) = E\psi(x)$$

where $H_0 = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \bar{V}$, $\Delta V = V(x) - \bar{V}$.

$$H_0\psi_k^0 = E_k^0\psi_k^0, \quad H_0\psi_{k'}^0 = E_{k'}^0\psi_{k'}^0$$

$$a(E_k^0 - E + \Delta V)\psi_k^0 + b(E_{k'}^0 - E + \Delta V)\psi_{k'}^0 = 0$$

$$\begin{cases} (E_k^0 - E)a + V_n^* b = 0, \\ V_n^a (E_{k'}^0 - E)b = 0 \end{cases}$$

$$V_n = \langle k|V(x)|k\rangle = \frac{1}{a}\int_0^a U_0 \delta(x)\, \mathrm{e}^{-\mathrm{i}2\pi m x/a}\, \mathrm{d}x$$

The Determine should be zero

$$\begin{vmatrix} E_k^0 - E & V_m^* \\ V_n & E_{k'}^0 - E \end{vmatrix} = 0$$

$E_k^0 = E_{k'}^0$,

$$E_\pm = \frac{1}{2}\{E_k^0 + E_{k'}^0 \pm \sqrt{(E_k^0 - E_k^0) + 4|V_n|^2}\}, \quad E_{\text{gap}} = 2|V_n| = \frac{2U_0}{a}$$

Solution. Near the Bragg points, assume

$$ka = n\pi + \kappa a, \quad \text{and} \quad qa = q_n a + \delta.$$

with $|\kappa|, |\delta| \ll 1$, $q_n = n\pi/a$. Keep the 2nd order of κ and δ, the implict equation can be written as

$$(-1)^n\left[1 - \frac{1}{2}(\kappa a)^2\right] = (-1)^n\left(1 - \frac{1}{2}\delta^2\right) + u\frac{(-1)^n\delta}{n\pi}.$$

Then, we can have the solution of δ

$$\delta = \frac{u}{n\pi} \pm \sqrt{\left(\frac{u}{n\pi}\right)^2 + (\kappa a)^2}.$$

Since $q = \sqrt{2mE}/\hbar$, then we have

$$E = \frac{\hbar^2(q_n + \delta/a)^2}{2m} \approx E_n^0 + \frac{\hbar^2 q_n \delta/a}{m},$$

where $E_n^0 = \frac{\hbar^2 q_n^2}{2m}$. Substitute $q_n = n\pi/a$, and the solution of δ into the expression of E

$$E = E_n^0 + \frac{n\pi\hbar^2\delta}{m} = E_n^0 + \frac{n\pi\hbar^2}{ma^2}\left[\frac{u}{n\pi} \pm \sqrt{\left(\frac{u}{n\pi}\right)^2 + (\kappa a)^2}\right].$$

Substitute $u = mU_0 a/\hbar^2$ into the expression above

$$E = E_n^0 + \frac{U_0}{a} \pm \sqrt{\left(\frac{U_0}{a}\right)^2 + \left(\frac{n\pi\hbar^2\kappa}{ma}\right)^2},$$

with the band gap

$$\Delta E_n = E_+ - E_- = 2\sqrt{\left(\frac{U_0}{a}\right)^2 + \left(\frac{n\pi\hbar^2\kappa}{ma}\right)^2}.$$

When $\kappa a \to 0$, the energy band and the band gap become

$$E = E_n^0 + \frac{U_0}{a} \pm \frac{U_0}{a}, \quad \text{and} \quad \Delta E_n = \frac{2U_0}{a}.$$

The results perfectly match that of the nearly-free electron model.

Problem 5.2 (Nearly free electron model). Consider 2D electrons subject to a weak periodic potential

$$U(x,y) = U_0\left(\cos\frac{2\pi x}{a} + \cos\frac{2\pi y}{a}\right).$$

(a) Find the energy bands near the edges of the first Brillouin zone. (Hint: the edges of the Brillouin zone are the Bragg planes where the eigenstates are doubly degenerate.) Plot the bands and isoenergetic surfaces.

(b) Repeat the analysis of part (a) for regions near the corners of the first Brillouin zone, where there are four degenerate eigenstates.

Solution (By TA). (a)

$$U(\boldsymbol{r}) = \sum_G U_G\,\mathrm{e}^{\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}, \quad U_G = \frac{1}{V_{\text{Cell}}}\int_{\text{Cell}} U(\boldsymbol{r})\,\mathrm{e}^{-\mathrm{i}\boldsymbol{G}\cdot\boldsymbol{r}}\,\mathrm{d}\boldsymbol{r}$$

$$U = U_0\left(\cos\frac{2\pi x}{a} + \cos\frac{2\pi y}{a}\right) = \frac{U_0}{2}(\mathrm{e}^{\pm\mathrm{i}\frac{2\pi x}{a}} + \mathrm{e}^{\pm\mathrm{i}\frac{2\pi x}{a}})$$

$$\boldsymbol{G} = (\pm\frac{2\pi}{a}, 0), (0, \pm\frac{2\pi}{a}), U_G = \frac{U_0}{2}.$$

$k_x = \frac{\pi}{a} + \delta$, k_y.

$$\begin{vmatrix} E_k^0 - E & \frac{U_0}{2} \\ \frac{U_0}{2} & E_{k'}^0 - E \end{vmatrix} = 0$$

$$E = \frac{\hbar^2}{2m}\left(\frac{\pi^2}{a^2} + q_x^2 + k_y^2\right) \pm \sqrt{\left(\frac{\pi\hbar^2}{ma}q_x\right)^2 + \left(\frac{U_0}{2}\right)^2}$$

$$k_x = \pm\frac{\pi}{a} + q_x, \quad k_y = \pm\frac{\pi}{a} + q_x.$$

(b) $\langle k_i|U(\boldsymbol{r})|k_j\rangle \neq 0$, $\boldsymbol{k}_i - \boldsymbol{k}_j = (0, \pm\frac{2\pi}{a})$, $(\pm\frac{2\pi}{a}, 0)$.

Solution.

(a) In reciprocal space, the reciprocal lattice vectors are

$$\boldsymbol{G}_1 = \left(\frac{2\pi}{a}, 0\right), \quad \boldsymbol{G}_2 = \left(0, \frac{2\pi}{a}\right).$$

The potential can be expressed as

$$U(x,y) = \frac{U_0}{2}\Big[e^{i\boldsymbol{G}_1\cdot\boldsymbol{r}} + e^{-i\boldsymbol{G}_1\cdot\boldsymbol{r}} + e^{i\boldsymbol{G}_2\cdot\boldsymbol{r}} + e^{-i\boldsymbol{G}_2\cdot\boldsymbol{r}}\Big].$$

At the zone edge (e.g., $k_x = \pi/a$), consider a wavevector

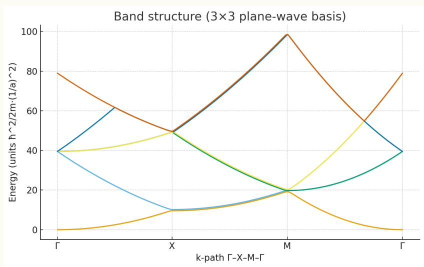
$$\boldsymbol{k} = \left(\frac{\pi}{a} + q_x, k_y\right),$$

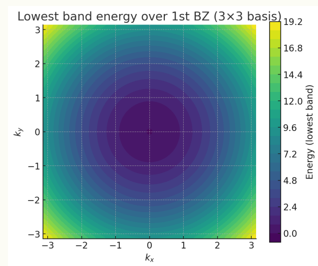
with $|q_x| \ll \pi/a$. The two nearly degenerate plane waves are $|\boldsymbol{k}\rangle$ and $|\boldsymbol{k}-\boldsymbol{G}_1\rangle$. Their free-electron energies are

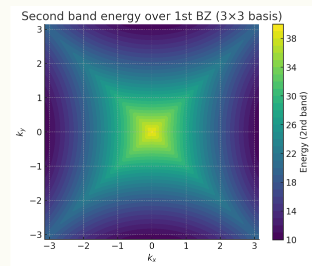
$$E^{(0)}(\boldsymbol{k}) = \frac{\hbar^2}{2m}\left[\left(\frac{\pi}{a}+q_x\right)^2 + k_y^2\right],$$
$$E^{(0)}(\boldsymbol{k}-\boldsymbol{G}_1) = \frac{\hbar^2}{2m}\left[\left(-\frac{\pi}{a}+q_x\right)^2 + k_y^2\right].$$

Expanding to first order in q_x

$$E^{(0)} \approx E_0 \pm \alpha q_x, \quad \text{where} \quad E_0 = \frac{\hbar^2}{2m}\left[\left(\frac{\pi}{a}\right)^2 + k_y^2\right], \quad \alpha = \frac{\hbar^2 \pi}{ma}.$$







The perturbation couples these states with matrix element $\langle \boldsymbol{k}|U|\boldsymbol{k}-\boldsymbol{G}_1\rangle = U_0/2$. The Hamiltonian in the subspace $\{|\boldsymbol{k}\rangle, |\boldsymbol{k}-\boldsymbol{G}_1\rangle\}$ is

$$H = \begin{pmatrix} E_0 + \alpha q_x & U_0/2 \\ U_0/2 & E_0 - \alpha q_x \end{pmatrix}.$$

Diagonalizing gives the eigenenergies

$$E_\pm = E_0 \pm \sqrt{(\alpha q_x)^2 + \left(\frac{U_0}{2}\right)^2} = E_0 + \frac{\hbar^2 q_x^2}{2m} \pm \sqrt{\left(\frac{\hbar^2\pi}{ma}q_x\right)^2 + \left(\frac{U_0}{2}\right)^2}$$

(b) The corners of the first Brillouin zone are points like $\boldsymbol{k}_0 = (\pi/a, \pi/a)$ (the M point). The four degenerate plane waves at this point are

$$\begin{aligned}
\boldsymbol{k}_1 &= \boldsymbol{k}_0 = (+\pi/a, +\pi/a), \\
\boldsymbol{k}_2 &= \boldsymbol{k}_0 - \boldsymbol{G}_1 = (-\pi/a, +\pi/a), \\
\boldsymbol{k}_3 &= \boldsymbol{k}_0 - \boldsymbol{G}_2 = (+\pi/a, -\pi/a), \\
\boldsymbol{k}_4 &= \boldsymbol{k}_0 - \boldsymbol{G}_1 - \boldsymbol{G}_2 = (-\pi/a, -\pi/a).
\end{aligned}$$

All have the same free-electron energy $E_0 = \frac{\hbar^2}{2m}[(\pi/a)^2 + (\pi/a)^2] = \frac{\hbar^2\pi^2}{ma^2}$.

In the basis $\{|\boldsymbol{k}_1\rangle, |\boldsymbol{k}_2\rangle, |\boldsymbol{k}_3\rangle, |\boldsymbol{k}_4\rangle\}$, states are coupled if they differ by $\pm\boldsymbol{G}_1$ or $\pm\boldsymbol{G}_2$. The non-zero off-diagonal matrix elements are

$$\langle \boldsymbol{k}_1|U|\boldsymbol{k}_2\rangle = \langle \boldsymbol{k}_1|U|\boldsymbol{k}_3\rangle = \langle \boldsymbol{k}_2|U|\boldsymbol{k}_4\rangle = \langle \boldsymbol{k}_3|U|\boldsymbol{k}_4\rangle = U_0/2.$$

The Hamiltonian is

$$H = E_0 \mathbb{1} + \frac{U_0}{2} \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}.$$

Diagonalizing this matrix yields eigenvalues

$$E = E_0 + \frac{U_0}{2} \times \{2,\ -2,\ 0,\ 0\} = \{E_0 + U_0,\ E_0 - U_0,\ E_0,\ E_0\}.$$

Thus, at $\boldsymbol{q} = 0$, the fourfold degeneracy splits into three distinct levels: one non-degenerate band at $E_0 + U_0$, one non-degenerate band at $E_0 - U_0$, and a two-fold degenerate band at E_0.

Lecture #6 Tight binding model

Problem 6.1 (Tight bindings in the BCC and FCC lattices)**.** We consider identical atoms located at points of a lattice that are body-centered cubic (**BCC: i**) and face-centered cubic (**FCC: ii**).

(a) For each case, determine the dispersion relation for s-electrons, having the general form

$$E = -\alpha - \gamma \sum_s \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_s},$$

where α and γ are the positive energies that can be calculated and $\mathbf{R}_s$ represents the vectors that connect an atom located at the origin to each of its nearest neighbors.

(b) Find the expressions for E at the points located at the intersection of the first Brillouin zone with the directions $[100]$, $[110]$, and $[111]$ and thus the value $E(\Gamma)$ to the center of this zone.

(c) Deduce the total width of the corresponding band, ΔE, and find the expression for the effective mass m^* in the vicinity of Γ as well as the shape of constant energy surfaces around this point.

Solution (By TA). $\frac{a}{2}(\pm1,\pm1,\pm1)$

$$\sum_s \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_s} = \sum_{u,v,w}\left[-\frac{a}{2}\mathrm{i}(uk_x+vk_y+wk_z)\right] = 8\cos\left(\frac{ak_x}{2}\right)\cos\left(\frac{ak_y}{2}\right)\cos\left(\frac{ak_z}{2}\right)$$

$\frac{a}{2}(\pm1,0,\pm1)$, $\frac{a}{2}(\pm1,\pm1,0)$, $\frac{a}{2}(0,\pm1,\pm1)$.
For BCC $E(\Gamma) = E(0,0,0) = -\alpha - 8\gamma$.

$$|\boldsymbol{k}|^2 = |\boldsymbol{k}-\boldsymbol{G}|^2 = |\boldsymbol{k}|^2 - 2\boldsymbol{k}\cdot\boldsymbol{G} + |\boldsymbol{G}|^2, \quad 2\boldsymbol{k}\cdot\boldsymbol{G} = |\boldsymbol{G}|^2$$

The nearest $\frac{2\pi}{a}(\pm1,0,\pm1)$

$$\left\{k_x \pm k_y = \pm\tfrac{2\pi}{a}, \quad k_y \pm k_z = \pm\tfrac{2\pi}{a}, \quad k_z \pm k_x = \pm\tfrac{2\pi}{a}.\right.$$

For $[1,0,0]$

$$\begin{cases} k_x = t, \\ k_y = k_z = 0 \end{cases}$$

$(\pm\frac{2\pi}{a},0,0)$, $E = -\alpha + 8\gamma$. For $[1,1,0]$

$$\begin{cases} k_x = k_y = t, \\ k_z = 0 \end{cases}$$

$(\frac{\pi}{a},\frac{\pi}{a},0)$, $E = -\alpha$. For $[1,1,1]$: $k_x = k_y = k_z = t$, $(\frac{\pi}{a},\frac{\pi}{a},\frac{\pi}{a})$, $E = -\alpha$.
For FCC:

$$E(\Gamma) = -\alpha - 12\gamma$$

Consider the nearest $\frac{2\pi}{a}(\pm1,\pm1,\pm1)$.

$$\boldsymbol{k}\cdot\boldsymbol{a} = \left(\frac{2\pi}{a}\right)^2 \cdot \frac{3}{2},$$

$$\pm k_x \pm k_y \pm k_z = \frac{3\pi}{a}$$

Sub-nearest: $\frac{2\pi}{a}(\pm 2, 0, 0)$,

$$k_x = \pm\frac{2\pi}{a}, \quad k_y = \pm\frac{2\pi}{a}, \quad k_z = \pm\frac{2\pi}{a}.$$

For $[1,0,0]$: $(\frac{2\pi}{a}, 0, 0)$

$$\begin{cases} k_x = t, \\ k_y = k_z = 0 \end{cases}, \quad E = -\alpha + 4r$$

For $[1,1,0]$: $(\frac{3\pi}{2a}, \frac{3\pi}{2a}, 0)$

$$\begin{cases} k_x = k_y = t, \\ k_z = 0 \end{cases}, \quad E = -\alpha - 2\gamma + 4\sqrt{2}r$$

For $[1,1,1]$: $(\frac{\pi}{a}, \frac{\pi}{a}, \frac{\pi}{a})$

$$k_x = k_y = k_z = t, \quad E = -\alpha$$

$$E_k = E_0 + \frac{\hbar^2}{2m}(\boldsymbol{k} - \boldsymbol{k}_0)^2 \quad \text{for } k \to 0$$

Use some approximation

$$k \to 0, \quad \cos\frac{ak}{2} = 1 - \frac{a^2k^2}{8}$$

BCC:

$$E_k = -\alpha - 8\gamma\left(1 - \frac{a^2k_x^2}{8}\right)(\ldots k_y)(\ldots k_z) = -\alpha - 8\gamma + \gamma a^2 k^2$$

where $\gamma a^2 = \frac{\hbar^2}{2m^*}$, $m^* = \frac{\hbar^2}{2\gamma a}$.

Solution (i = BCC, ii = FCC). Assume the lattice constant a.

($\mathrm{a_i}$) Each atom has 8 NNs, i.e.,

$$\{\boldsymbol{R}_s\} = \left\{\frac{a}{2}(\pm 1, \pm 1, \pm 1)\right\}$$

Then, the sum in the general form becomes

$$\sum_s \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_s} = \sum_{\sigma_x=\pm}\sum_{\sigma_y=\pm}\sum_{\sigma_z=\pm} \mathrm{e}^{-\mathrm{i}\frac{a}{2}\sigma_x k_x} \cdot \mathrm{e}^{-\mathrm{i}\frac{a}{2}\sigma_y k_y} \cdot \mathrm{e}^{-\mathrm{i}\frac{a}{2}\sigma_z k_z} = 8\cos\left(\frac{ak_x}{2}\right)\cos\left(\frac{ak_y}{2}\right)\cos\left(\frac{ak_z}{2}\right)$$

Hence, the dispersion relation is

$$E(\boldsymbol{k}) = -\alpha - 8\gamma\cos\left(\frac{ak_x}{2}\right)\cos\left(\frac{ak_y}{2}\right)\cos\left(\frac{ak_z}{2}\right)$$

($\mathrm{b_i}$)
i. Γ point: $\boldsymbol{k} = (0,0,0)$, $E(\Gamma) = -\alpha - 8\gamma$.
ii. $[100]$: $\boldsymbol{k} = \frac{2\pi}{a}(1,0,0)$, $E(X) = -\alpha + 8\gamma$.
iii. $[110]$: $\boldsymbol{k} = \frac{2\pi}{a}(\frac{1}{2}, \frac{1}{2}, 0)$, $E(N) = -\alpha$.
iv. $[111]$: $\boldsymbol{k} = \frac{\pi}{a}(1,1,1)$, $E(P) = -\alpha$

(c_i) The $\cos$ terms in the general form differs between $[-8\gamma, +8\gamma]$. So, the energy gap is

$$\Delta E_{\text{BCC}} = (-\alpha + 8\gamma) - (-\alpha - 8\gamma) = 16\gamma$$

The dispersion relation can be expanded around the Γ point

$$E(\boldsymbol{k}) = -\alpha - 8\gamma\left[1 - \frac{1}{2}\left(\frac{ak_x}{2}\right)^2 - \frac{1}{2}\left(\frac{ak_y}{2}\right)^2 - \frac{1}{2}\left(\frac{ak_z}{2}\right)^2\right] \xlongequal{k^2=k_x^2+k_y^2+k_z^2} -\alpha - 8\gamma + \gamma a^2 k^2$$

Then, the effective mass $m^* = \frac{\hbar^2}{\partial^2 E/\partial k^2} = \frac{\hbar^2}{2\gamma a^2}$.

(a_{ii}) Each atom has 12 NNs, i.e.,

$$\{\boldsymbol{R}_s\} = \left\{\frac{a}{2}(\pm1, \pm1, 0), \frac{a}{2}(\pm1, 0, \pm1), \frac{a}{2}(0, \pm1, \pm1).\right\}$$

Then, the sum in the general form becomes

$$\begin{aligned}\sum_s \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{R}_s} &= \sum_{\text{cyclic}(x,y,z)} \sum_{\pm} \mathrm{e}^{-\mathrm{i}\frac{a}{2}(\pm k_i \pm k_j)} \\ &= 4\left[\cos\left(\frac{ak_x}{2}\right)\cos\left(\frac{ak_y}{2}\right) + \cos\left(\frac{ak_y}{2}\right)\cos\left(\frac{ak_z}{2}\right) + \cos\left(\frac{ak_z}{2}\right)\cos\left(\frac{ak_x}{2}\right)\right]\end{aligned}$$

Hence, the dispersion relation is

$$E(\boldsymbol{k}) = -\alpha - 4\gamma\left[\cos\left(\frac{ak_x}{2}\right)\cos\left(\frac{ak_y}{2}\right) + \cos\left(\frac{ak_y}{2}\right)\cos\left(\frac{ak_z}{2}\right) + \cos\left(\frac{ak_z}{2}\right)\cos\left(\frac{ak_x}{2}\right)\right]$$

(b_{ii}) i. Γ point: $\boldsymbol{k} = (0,0,0)$, $E(\Gamma) = -\alpha - 8\gamma$. ii. $[100]$: $\boldsymbol{k} = \frac{2\pi}{a}(1,0,0)$, $E(X) = -\alpha - 4\gamma$.

iii. $[110]$ point: $\boldsymbol{k} = \frac{2\pi}{a}(\frac{3}{4}, \frac{3}{4}, 0)$, $E(N) = -\alpha + \gamma(4\sqrt{2} - 2)$.

iv. $[111]$ point: $\boldsymbol{k} = \frac{\pi}{a}(1,1,1)$, $E(P) = -\alpha - 12\gamma$

(c_{ii}) The $\cos$ terms in the general form differs between $[-12\gamma, +4\gamma]$. So, the energy gap is

$$\Delta E_{\text{FCC}} = (-\alpha + 4\gamma) - (-\alpha - 12\gamma) = 16\gamma$$

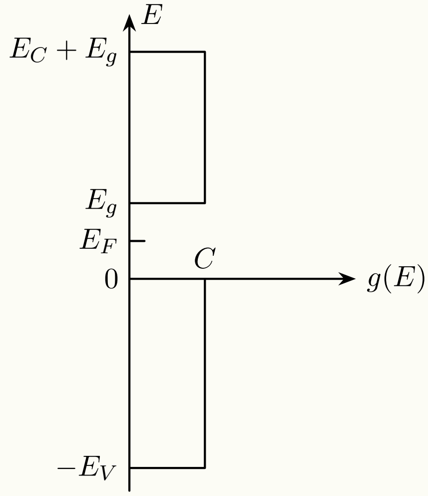
The dispersion relation can be expanded around the Γ point

$$\begin{aligned}E(\boldsymbol{k}) &= -\alpha - 4\gamma \sum_{\text{cyclic}(x,y,z)} \left[1 - \frac{1}{2}\left(\frac{ak_i}{2}\right)^2 - \frac{1}{2}\left(\frac{ak_j}{2}\right)^2\right] \\ &= -\alpha - 12\gamma + 4\gamma\left[\left(\frac{k_x a}{2}\right)^2 + \left(\frac{k_y a}{2}\right)^2 + \left(\frac{k_z a}{2}\right)^2\right] = -\alpha - 12\gamma + \gamma a^2 k^2\end{aligned}$$

Then, the effective mass $m^* = \frac{\hbar^2}{\partial^2 E/\partial k^2} = \frac{\hbar^2}{2\gamma a^2}$.

Lecture #7 Semiconductor

Problem 7.1 (Elementary study of an intrinsic semiconductor). Consider a semiconductor characterized by a constant density of states ($g(E) = C$) in both the valence and conduction band. These bands, separated by E_g, have a respective width of E_V and E_C (see the Figure).



(a) Knowing that the Fermi level E_F is in the bandgap ($5k_BT < E_F < E_g - 5k_BT$), leading to the conventional simplification for $f(E)$ (The Boltzmann approximation can be applied), find the expression for the number of electrons n in the conduction band and the number of holes h in the valence band respectively as a function of the data and absolute temperature.

(b) Deduce the expression for the volumetric density of intrinsic carriers n_i at ambient temperature ($k_BT = 25\,\mathrm{meV}$) as well as the position of the Fermi level E_{F_i}.

Numerical applications: $E_g = 0.7\,\mathrm{eV}$, $E_C = E_V = 5\,\mathrm{eV}$, and $C = 2 \times 10^{21}\,\mathrm{electrons/cm^3/eV}$.

(c) At the same temperature, what is the intrinsic electronic conductivity of this material? Take $\mu_h = \mu_e = 1000\,\mathrm{cm^2/Vs}$.

(d) The material is doped with N_0 boron impurities. Find the expressions and the new values of n_d, h_d, σ_d, and E_{F_d} with $N_0 = 10^{12}$ impurities per cm^3 and next with $N_0 = 10^{14}$ impurities per cm^3.

Solution.

(a) The number of electrons in the conduction band

$$n = \int_{E_g}^{E_g+E_c} g(E)f(E)\,\mathrm{d}E = Ck_BT\,\mathrm{e}^{-\frac{E_g-E_F}{k_BT}}(1-\mathrm{e}^{-\frac{E_C}{k_BT}}) \xlongequal[\text{Boltzmann approximation}]{E_C \gg k_BT} Ck_BT\,\mathrm{e}^{-\frac{E_g-E_F}{k_BT}}.$$

The number of holes in the valence band

$$h = \int_{-E_V}^{0} g(E)[1-f(E)]\,\mathrm{d}E \approxeq Ck_BT\,\mathrm{e}^{-\frac{E_F}{k_BT}}(1-\mathrm{e}^{-\frac{E_V}{k_BT}}) \xlongequal[\text{Boltzmann approximation}]{E_V \gg k_BT} Ck_BT\,\mathrm{e}^{-\frac{E_F}{k_BT}}.$$

where $f(E) \approxeq \mathrm{e}^{-\frac{E-E_F}{k_BT}}$, $[1-f(E)] \approxeq \mathrm{e}^{\frac{E-E_F}{k_BT}}$.

(b) For the intrinsic carriers, the number of electrons and holes are the same

$$n = Ck_BT\,\mathrm{e}^{-\frac{E_g-E_F}{k_BT}} = h = Ck_BT\,\mathrm{e}^{-\frac{E_F}{k_BT}}, \quad E_g - E_F = E_F.$$

Then, the Fermi level is at mid-gap, i.e., $E_F = \frac{E_g}{2}$. Substitute it back into the expression of n (or h), we get the intrinsic carrier concentration

$$n_i = Ck_BT\,\mathrm{e}^{-\frac{E_g}{2k_BT}}.$$

Apply the parameters, $n_i = 4.158 \times 10^{19}\,\mathrm{m}^{-3}$, $E_F = 0.35\,\mathrm{eV}$.

(c) The conductivity

$$\sigma_i = e(n_i\mu_e + n_i\mu_h) = 1.332\,\mathrm{S/m}.$$

(d) Boron is an acceptor, so doping concentration $N_A = N_0$. Now, the number of electrons and holes

$$n = n_i,\ h = n_i \longrightarrow n_d = n,\ h_d = n + N_0,$$

and they satisfy the identity: mass-action law

$$n_i^2 = n_d h_d = n(n + N_0).$$

We can solve that

$$n = \frac{\sqrt{N_0^2 + 4n_i^2} - N_0}{2}, \quad n_d = n = \frac{\sqrt{N_0^2 + 4n_i^2} - N_0}{2}, \quad h_d = n + N_0 = \frac{\sqrt{N_0^2 + 4n_i^2} + N_0}{2}.$$

The conductivity becomes

$$\sigma_d = e(n_d\mu_e + h_d\mu_h).$$

From $h = Ck_BT\,\mathrm{e}^{-\frac{E_F}{k_BT}}$, we can obtain the Fermi level

$$E_{F_d} = k_BT \ln \frac{Ck_BT}{h_d}.$$

Apply the parameters (with the previous $n_i = 4.158 \times 10^{19}\,\mathrm{m}^{-3}$)

i. $N_0 = 10^{18}\,\mathrm{m}^{-3}$.
The numbers of the acceptors and donators

$$n_d = 4.108 \times 10^{19}\,\mathrm{m}^{-3},$$
$$h_d = 4.208 \times 10^{19}\,\mathrm{m}^{-3}.$$

The conductivity and the Fermi level

$$\sigma_d = 1.332\,\mathrm{S/m},\ E_{F_d} = 0.350\,\mathrm{eV}.$$

ii. $N_0 = 10^{20}\,\mathrm{m}^{-3}$.
The numbers of the acceptors and donators

$$n_d = 1.503 \times 10^{19}\,\mathrm{m}^{-3},$$
$$h_d = 1.150 \times 10^{20}\,\mathrm{m}^{-3}.$$

The conductivity and the Fermi level

$$\sigma_d = 2.083\,\mathrm{S/m},\ E_{F_d} = 0.325\,\mathrm{eV}.$$

Lecture #8 Electrons in a magnetic field

Problem 8.1 (Onsager formula)**.** Suppose that the electron spectrum for a tetragonal lattice can be described by the following equation

$$E_k = \frac{\hbar^2 k_x^2}{2m_{xx}} + \frac{\hbar^2 k_y^2}{2m_{xx}} + \frac{\hbar^2 k_z^2}{2m_{zz}},$$

with $m_{xx} \neq m_{zz}$. Using the Onsager formula for the cyclotron mass, $m_c = \frac{\hbar^2}{2\pi}\frac{\partial A}{\partial E_k}$ with A being the area of an isoenergetic contour perpendicular to the magnetic field. Find m_c for the case when the magnetic field is

(a) Along the z-axis.

(b) In the $x - y$ plane.

Solution.

(a) For $\boldsymbol{B} \parallel \hat{z}$, the motion of electrons will perpendicular to $\hat{z}$, i.e., parallel to $x - y$ plane. The equation of motion becomes

$$E_k = \frac{\hbar^2}{2m_{xx}}(k_x^2 + k_y^2),$$

with the equation of the motion boundary in k-space

$$k_\parallel^2 \equiv k_x^2 + k_y^2 = \frac{2m_{xx}E_k}{\hbar^2}.$$

Then, we get the radius. The circle area of the isoenergetic contour is

$$A = \pi k_\parallel^2 = \frac{2\pi m_{xx}E_k}{\hbar^2}.$$

Substituting to the definition of cyclotron mass, we can obtain

$$m_c = \frac{\hbar^2}{2\pi}\frac{\partial A}{\partial E_k} = \frac{\hbar^2}{2\pi}\frac{2\pi m_{xx}}{\hbar^2} = m_{xx}.$$

(b) Simply taking $\boldsymbol{B} \parallel \hat{x}$, then orbits lie in the $k_y - k_z$ plane with an ellipse area. The equation of motion becomes

$$E_k = \frac{\hbar^2 k_y^2}{2m_{xx}} + \frac{\hbar^2 k_z^2}{2m_{zz}},$$

with the equation of the motion boundary in k-space

$$\frac{k_y^2}{2m_{xx}E/\hbar^2} + \frac{k_z^2}{2m_{zz}E/\hbar^2} = 1.$$

Then we get the semi-axes. The ellipse area of the isoenergetic contour is

$$A = \pi\sqrt{\frac{2m_{xx}E}{\hbar^2}}\sqrt{\frac{2m_{zz}E}{\hbar^2}} = \frac{2\pi E}{\hbar^2}\sqrt{m_{xx}m_{zz}}$$

Substituting to the definition of cyclotron mass, we can obtain

$$m_c = \frac{\hbar^2}{2\pi}\frac{\partial A}{\partial E_k} = \sqrt{m_{xx}m_{zz}}.$$

Problem 8.2 (Integer Quantum Hall Effect)**.** The Hall resistance R_{xy} of a two-dimensional electron gas is measured as a function of magnetic field B at a very low temperature. A series of plateaus are observed at values $R_{xy} = \frac{h}{(\nu e^2)}$, where ν is an integer. Simultaneously, the longitudinal resistance R_{xx} shows peaks and drops to zero.

(a) Explain physically why R_{xx} drops to zero whenever R_{xy} is on a plateau.

(b) A sample has an electron density of $n_s = 4.0 \times 10^{11}\,\mathrm{cm}^{-2}$.

  i. Calculate the filling factor ν at $B = 2.0\,\mathrm{T}$.

  ii. What is the expected value of R_{xy} on the plateau nearest to this field?

  iii. How many Landau levels are completely filled at this plateau?

(c) Why is the IQHE used as a resistance standard? Which two fundamental constants are involved?

Solution.

(a) When the Hall resistance R_{xy} is on a plateau, the Fermi energy lies in a mobility gap between Landau levels. All extended states in the bulk are absent, making the bulk insulating. Current flows only along dissipationless chiral edge states.

At the edges, backscattering is prevented by spatial separation: even if impurities or imperfections exist along an edge, they may cause small forward deviations in an electron's path but cannot reverse its direction, so no resistance is produced. With no extended states in the bulk to allow dissipation and no backscattering at the edges, the longitudinal voltage drop vanishes, giving $R_{xx} = 0$.

(b) i. From the degeneracy of Landau level $n_s = \nu \frac{eB}{h}$, we can obtain the filling factor

$$\nu = \frac{n_s h}{eB} = \frac{4.0 \times 10^{15}\,\mathrm{m}^{-2} \times 6.626 \times 10^{-34}\,\mathrm{J \cdot s}}{1.602 \times 10^{-19}\,\mathrm{C} \times 2.0\,\mathrm{T}} = 8.271.$$

  ii. Substituting the values into the expression of the Hall resistance

$$R_{xy} = \frac{R_K}{\tilde{\nu}} = \frac{h}{\tilde{\nu} e^2} = 3226.60\,\Omega.$$

  where $\tilde{\nu}$ is the nearest integer of ν.

  iii. The filling factor $\tilde{\nu} = 8$ directly corresponds to the number of completely filled Landau levels.

(c) IQHE provides a quantized Hall resistance

$$R_H = \frac{h}{e^2},$$

which precisely quantized and only based on two fundamental constants: The Planck constant h and the elementary charge e, so it is used as a resistance standard.

Problem 8.3 (Lifshitz-Kosevich formula). The Lifshitz-Kosevich formula describes the amplitude of the Shubnikov-de Haas (SdH) oscillations. The amplitude ΔR is proportional to the factor: $R_T = \frac{\lambda(T)}{\operatorname{sh}\lambda(T)}$, where $\lambda(T) = 2\pi^2 \frac{k_B T m^*}{\hbar e B}$.

(a) A researcher measures the SdH oscillation amplitude ΔR at a fixed magnetic field B as a function of temperature T. How can they use this data to determine the effective mass m^* of the charge carriers?

(b) Another researcher fixes the temperature T and measures the SdH amplitude ΔR as a function of $1/B$. They find that the amplitude decreases exponentially with $1/B$. What physical parameter does this decay reveal, and what does a rapid decay imply about the quality of the sample?

Solution.

(a) Take $\alpha = \frac{2\pi^2 k_B m^*}{\hbar e B}$, then $\lambda(T) = \alpha T$. Since at $T \gg 1$, the amplitude ΔR satisfies

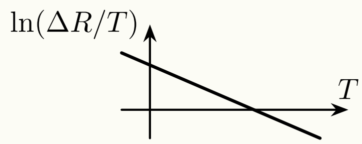
$$\Delta R \propto R_T = \frac{\alpha T}{\operatorname{sh}(\alpha T)} \to \frac{1}{2}\alpha T\, \mathrm{e}^{-\alpha T}, \quad \Delta R = \tilde{C} T\, \mathrm{e}^{-\alpha T}$$

Take the logarithm to the two sides

$$\ln \Delta R = \ln \tilde{C} + \ln T - \alpha T, \quad \ln\left(\frac{\Delta R}{T}\right) = -\alpha T + C,$$

where $C = \ln \tilde{C}$.

So, one can take the data at high temperature, and use some software tool to convert the original data from T-ΔR to T-$\ln\left(\frac{\Delta R}{T}\right)$, then plot the new data set like the right figure.



By fitting the slope $k = -\alpha$ that fits the figure, one can obtain the effective mass

$$m^* = \frac{\alpha \hbar e B}{2\pi^2 k_B} = -\frac{k \hbar e B}{2\pi^2 k_B}$$

(b) Similarly to part (a), we take $T \gg 1$. Since ΔR is proportional to $R_T R_D R_S$, and if we only consider the magnetic variable B, then $\Delta R \propto R_T R_D$. To be more specific,

$$\Delta R \propto \frac{2\pi^2 k_B T m^*}{\hbar e} \cdot \frac{1}{B} \cdot \mathrm{e}^{-\frac{2\pi^2 k_B T m^*}{\hbar e} \cdot \frac{1}{B}} \cdot \mathrm{e}^{-\frac{\pi m^*}{e \tau_q} \cdot \frac{1}{B}} = \tilde{C} \cdot \frac{1}{B} \cdot \mathrm{e}^{-(\alpha + \beta/\tau_q) \cdot \frac{1}{B}}$$

where the coefficients α and β can be easily known since we have already obtained the effective mass. Taking the logarithm to both sides, we can obtain

$$\ln(\Delta R) = C - \ln B - \left(\alpha + \frac{\beta}{\tau_q}\right)\frac{1}{B},$$

Since the effective mass m^* has been determined from (a), so the coefficients α and β are known, and the slope directly reveals the quantum scattering rate $1/\tau_q$.

A rapid exponential decay corresponds to a short τ_q, meaning a short mean free path and therefore significant disorder: an indicator of poor sample quality.

Lecture #9 Band structure of graphene

Problem 9.1. In the calculation of the monolayer graphene band structure using tight-binding model, if we keep the contribution of the next-nearest hopping t'.

(a) Write down the dispersion relation.

(b) Expand the energy spectrum around the Dirac point up to second order. Search the literature and check what's trigonal warping effect is in a monolayer.

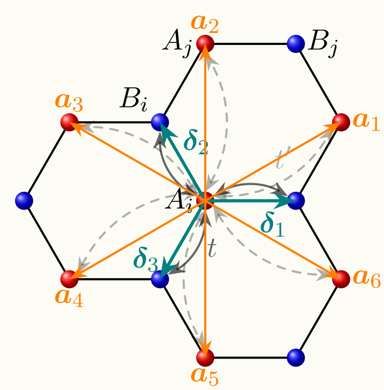
Solution.

(a) The Hamiltonian in second quantization form for the graphene with the next-nearest (NN) hopping is

$$\mathcal{H} = -t\sum_{\langle i,j\rangle}(\hat{a}_i\hat{b}_j^\dagger + \text{H.c.}) - t'\sum_{\langle\langle i,j\rangle\rangle}(\hat{a}_i\hat{a}_j^\dagger + \hat{b}_i\hat{b}_j^\dagger + \text{H.c.}) = \mathcal{H}_t + \mathcal{H}_{t'},$$

where i and j label the nearest neighbor, i.e., the bond A_i, B_i.

The fermion operators a_i and $a_i^\dagger$ annihilate/create an electron at site A, also for b_i and $b_i^\dagger$ at site B. The vectors pointing from A_i to the 3 nearest B-type sites and 6 next-nearest A-type sites are



$$\begin{aligned}
&\boldsymbol{\delta}_1 = a(1,0), &&\boldsymbol{\delta}_2 = a\left(-\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}\right), &&\boldsymbol{\delta}_3 = a\left(-\tfrac{1}{2}, -\tfrac{\sqrt{3}}{2}\right);\\
&\boldsymbol{a}_1 = a\left(\tfrac{3}{2}, \tfrac{\sqrt{3}}{2}\right), &&\boldsymbol{a}_2 = a(0,\sqrt{3}), &&\boldsymbol{a}_3 = a\left(-\tfrac{3}{2}, \tfrac{\sqrt{3}}{2}\right),\\
&\boldsymbol{a}_4 = a\left(-\tfrac{3}{2}, -\tfrac{\sqrt{3}}{2}\right), &&\boldsymbol{a}_5 = a(0,-\sqrt{3}), &&\boldsymbol{a}_6 = a\left(\tfrac{3}{2}, -\tfrac{\sqrt{3}}{2}\right).
\end{aligned}$$

Assume there are N atoms in total, then there are $N/2$ A-type or B-type sites in total.

Applying the Fourier transformation to the fermion operators

$$\begin{aligned}
\hat{a}_i &= \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}_i}\hat{a}_{\boldsymbol{k}}, \qquad \hat{b}_{i+\delta} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k},j} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{r}_i+\boldsymbol{\delta}_j)}\hat{b}_{\boldsymbol{k}},\\
\hat{a}_i^\dagger &= \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}_i}\hat{a}_{\boldsymbol{k}}^\dagger, \quad \hat{b}_{i+\delta}^\dagger = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k},j} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{r}_i+\boldsymbol{\delta}_j)}\hat{b}_{\boldsymbol{k}}^\dagger.
\end{aligned}$$

Consider the first term of the Hamiltonian, it can be expressed as

$$\begin{aligned}
\mathcal{H}_t &= -t\sum_{i\in A}\sum_{j=1}^{3}\left(\frac{1}{N}\sum_{\boldsymbol{k}}\mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{r}_i}\hat{a}_{\boldsymbol{k}}^\dagger\sum_{\boldsymbol{k}'}\mathrm{e}^{\mathrm{i}\boldsymbol{k}'\cdot(\boldsymbol{r}_i+\boldsymbol{\delta}_j)}\hat{b}_{\boldsymbol{k}'} + \text{h.c.}\right)\\
&= -t\frac{1}{N}\sum_{\boldsymbol{k},\boldsymbol{k}'}\sum_{i\in A}\mathrm{e}^{\mathrm{i}(\boldsymbol{k}'-\boldsymbol{k})\cdot\boldsymbol{r}_i}\sum_{j=1}^{3}\mathrm{e}^{\mathrm{i}\boldsymbol{k}'\cdot\boldsymbol{\delta}_j}\hat{a}_{\boldsymbol{k}}^\dagger\hat{b}_{\boldsymbol{k}'} + \text{h.c.} = -t\sum_{\boldsymbol{k}}\sum_{j=1}^{3}\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}_j}\hat{a}_{\boldsymbol{k}}^\dagger\hat{b}_{\boldsymbol{k}} + \text{h.c.}
\end{aligned}$$

where $\sum_{i\in A}\mathrm{e}^{\mathrm{i}(\boldsymbol{k}'-\boldsymbol{k})\cdot\boldsymbol{r}_i} = N\delta_{\boldsymbol{k}',\boldsymbol{k}}$ is applied, and multiple sum will not occur here since we only sum over every A-type site.

Denote $\Delta(\boldsymbol{k}) = -t\sum_{j=1}^{3} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{\delta}_j}$, this part becomes

$$\mathcal{H}_t = \sum_{\boldsymbol{k}} \Delta(\boldsymbol{k})\hat{a}_{\boldsymbol{k}}^\dagger \hat{b}_{\boldsymbol{k}} + \sum_{\boldsymbol{k}} \Delta^*(\boldsymbol{k})\hat{b}_{\boldsymbol{k}}^\dagger \hat{a}_{\boldsymbol{k}}$$

Similarly for the second part: Denote $\epsilon(\boldsymbol{k}) = -t'\sum_{m=1}^{6} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{a}_m}$, one can obtain (The Hermit conjugate is applied, but the factor $\frac{1}{2}$ is also applied to prevent a twice sum between the same NN bond.)

$$\mathcal{H}_{t'} = \sum_{\boldsymbol{k}} \epsilon(\boldsymbol{k})\hat{a}_{\boldsymbol{k}}^\dagger \hat{a}_{\boldsymbol{k}} + \sum_{\boldsymbol{k}} \epsilon(\boldsymbol{k})\hat{b}_{\boldsymbol{k}}^\dagger \hat{b}_{\boldsymbol{k}}.$$

Here the $\epsilon(\boldsymbol{k})$ should not taken Hermit conjugate, since the hopping A_i-A_j and B_i-B_j are the same physical process. Hence, the total Hamiltonian is

$$\begin{aligned}\mathcal{H} = \mathcal{H}_t + \mathcal{H}_{t'} = \sum_{\boldsymbol{k}}\Big(\Delta(\boldsymbol{k})\hat{a}_{\boldsymbol{k}}^\dagger \hat{b}_{\boldsymbol{k}} + \Delta^*(\boldsymbol{k})\hat{b}_{\boldsymbol{k}}^\dagger \hat{a}_{\boldsymbol{k}} + \epsilon(\boldsymbol{k})\hat{a}_{\boldsymbol{k}}^\dagger \hat{a}_{\boldsymbol{k}} + \epsilon^*(\boldsymbol{k})\hat{b}_{\boldsymbol{k}}^\dagger \hat{b}_{\boldsymbol{k}}\Big)&\\ = \sum_{\boldsymbol{k}}\Big(\hat{a}_{\boldsymbol{k}}^\dagger \hat{b}_{\boldsymbol{k}}^\dagger\Big)\begin{pmatrix}\epsilon(\boldsymbol{k}) & \Delta(\boldsymbol{k})\\ \Delta^*(\boldsymbol{k}) & \epsilon(\boldsymbol{k})\end{pmatrix}\begin{pmatrix}\hat{a}_{\boldsymbol{k}}\\ \hat{b}_{\boldsymbol{k}}\end{pmatrix} = \sum_{\boldsymbol{k}} \boldsymbol{\Psi}^\dagger \mathbf{H}_{2\times 2}\boldsymbol{\Psi}&\end{aligned}$$

By taking $\det(\mathbf{H}_{2\times 2} - E\mathbb{1}) = 0$, one can obtain the dispersion relation

$$(\epsilon(\boldsymbol{k}) - E)^2 = \Delta(\boldsymbol{k})\Delta^*(\boldsymbol{k}), \quad E_\pm = \pm\sqrt{\Delta(\boldsymbol{k})\Delta^*(\boldsymbol{k})} + \epsilon(\boldsymbol{k})$$

To obtain the exact expression, expanding $\Delta(\boldsymbol{k})$ and $\epsilon(\boldsymbol{k})$

$$\begin{aligned}
\Delta(\boldsymbol{k}) \quad &= -t\Big[\mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot a(1,0)} + \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot a\left(-\frac{1}{2},\frac{\sqrt{3}}{2}\right)} + \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot a\left(-\frac{1}{2},-\frac{\sqrt{3}}{2}\right)}\Big]\\
&= -t\Big(\mathrm{e}^{\mathrm{i}k_x a} + \mathrm{e}^{-\frac{1}{2}\mathrm{i}k_x a}\cdot \mathrm{e}^{\frac{\sqrt{3}}{2}\mathrm{i}k_y a} + \mathrm{e}^{-\frac{1}{2}\mathrm{i}k_x a}\cdot \mathrm{e}^{-\frac{\sqrt{3}}{2}\mathrm{i}k_y a}\Big)\\
&= -t\,\mathrm{e}^{-\mathrm{i}k_x a}\Big[1 + 2\,\mathrm{e}^{\frac{3}{2}\mathrm{i}k_x a}\cos\Big(\tfrac{\sqrt{3}}{2}k_y a\Big)\Big],\\
\Delta(\boldsymbol{k})\Delta^*(\boldsymbol{k}) &= t^2\Big[3 + \underbrace{2\cos(\sqrt{3}k_y a) + 4\cos\Big(\frac{3}{2}k_x a\Big)\cos\Big(\frac{\sqrt{3}}{2}k_y a\Big)}_{f(\boldsymbol{k})}\Big],\\
\epsilon(\boldsymbol{k}) \quad &= -t'\Big[\mathrm{e}^{\mathrm{i}ka\left(\frac{3}{2},\frac{\sqrt{3}}{2}\right)} + \mathrm{e}^{\mathrm{i}ka(0,\sqrt{3})} + \mathrm{e}^{\mathrm{i}ka\left(-\frac{3}{2},\frac{\sqrt{3}}{2}\right)}\\
&\quad + \mathrm{e}^{\mathrm{i}ka\left(-\frac{3}{2},-\frac{\sqrt{3}}{2}\right)} + \mathrm{e}^{\mathrm{i}ka(0,-\sqrt{3})} + \mathrm{e}^{\mathrm{i}ka\left(\frac{3}{2},-\frac{\sqrt{3}}{2}\right)}\Big]\\
&= -t'\Big[\mathrm{e}^{\sqrt{3}\mathrm{i}k_y a} + 2\,\mathrm{e}^{\frac{\sqrt{3}}{2}\mathrm{i}k_y a}\cos\Big(\tfrac{3}{2}k_x a\Big) + 2\,\mathrm{e}^{-\frac{\sqrt{3}}{2}\mathrm{i}k_y a}\cos\Big(\tfrac{3}{2}k_x a\Big) + \mathrm{e}^{-\sqrt{3}\mathrm{i}k_y a}\Big]\\
&= -t'\Big[2\cos(\sqrt{3}k_x a) + 4\cos\Big(\tfrac{\sqrt{3}}{2}k_y a\Big)\cos\Big(\tfrac{3}{2}k_x a\Big)\Big] = -t' f(\boldsymbol{k})
\end{aligned}$$

Substituting them back to the dispersion relation, we can obtain

$$E_\pm = \pm t\sqrt{3 + f(\boldsymbol{k})} - t' f(\boldsymbol{k})$$

with $f(\boldsymbol{k})$ defined above.

(b) Take the Dirac point $\boldsymbol{K} = \frac{2\pi}{3\sqrt{3}a}(\sqrt{3}, 1)$, then calculate $f(\boldsymbol{K}+\boldsymbol{q})$ ($|\boldsymbol{q}| \ll 1$).

$$f(\boldsymbol{K}+\boldsymbol{q}) = 2\cos(\sqrt{3}K_y a + \sqrt{3}q_y a) + 4\cos\left(\frac{3}{2}K_x a + \frac{3}{2}q_x a\right)\cos\left(\frac{\sqrt{3}}{2}K_y a + \frac{\sqrt{3}}{2}q_y a\right) = -3 + \frac{9}{4}|\boldsymbol{q}|^2 a^2,$$

where the identities for the trigonometric functions and the approximations $\cos(x) \sim 1 - \frac{1}{2}x^2$, $\sin x \sim x$ are used. Substituting $f(\boldsymbol{K}+\boldsymbol{q})$ to $E_\pm$

$$E_\pm = \pm t\sqrt{3 - 3 + \frac{9}{4}|\boldsymbol{q}|^2 a^2} - t'\left(-3 + \frac{9}{4}|\boldsymbol{q}|^2 a^2\right) + \mathcal{O}(|\boldsymbol{q}|^3) = \pm\frac{3t}{2}|\boldsymbol{q}|a + 3t' - \frac{9}{4}t'|\boldsymbol{q}|^2 a^2 + \mathcal{O}(|\boldsymbol{q}|^3).$$

Problem 9.2. Graphene has the record room-temperature electron mobility, mainly due to two reasons. The first is the high Debye temperature ($\sim 2100\,\mathrm{K}$) that suppresses phonon scattering arising from the in-plane covalent sp^2 bonding between carbon atoms. The second one is related to its unique linear Dirac band, resulting in massless Dirac fermions ($m^* = 0$) near the Dirac point. However, if we apply the definition of the effective mass $m^* = \hbar^2\left(\frac{\mathrm{d}^2E(k)}{\mathrm{d}k^2}\right)^{-1}$ to the band dispersion of graphene ($E_\pm(k) = \pm\hbar v_F k$), we will get diverging m^* and non-differentiable Dirac point. Please explain this contradiction.

Solution. The contradiction arises from applying different definitions of “mass” to graphene’s unique band structure. The description of “massless Dirac fermions” comes from the analogy to relativistic particles.

From a massless special relativity particle like a photon to the electrons in the graphene, their dispersion relations have the comparasion

$$E = pc \Longrightarrow E = \pm\hbar v_F k = p v_F$$

with v_F playing the role of c. Hence, electrons near the Dirac point behave as if they have zero rest mass.

However, the conventional effective mass $m^* = \hbar^2(\mathrm{d}^2E/\mathrm{d}k^2)^{-1}$ is defined under the parabolic-band approximation $E \propto k^2$. For graphene’s linear dispersion, $\mathrm{d}^2E/\mathrm{d}k^2 = 0$ away from $k = 0$, leading to $m^* \to \infty$ mathematically. At $k = 0$ the dispersion is non-differentiable, so the formula breaks down.

Physically, this infinite effective mass reflects that the electron’s speed is constant at v_F: An external force parallel to its momentum cannot change its speed, much like a photon’s speed is fixed at c. Thus, the “massless” label refers to the relativistic analogy, while the divergent m^* indicates the inapplicability of the parabolic effective mass concept to a linear band.